Playtime–wellbeing associations are practically null across video game genres

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Video game research often treats games as a monolithic category, and single-game or single-platform studies leave open whether findings generalise across the medium. In this registered report, we used multi-platform digital trace data from the Open Play dataset to test whether (H1) total playtime is meaningfully associated with mental wellbeing and (H2) whether this association differs by game genre. The data comprise 226,000 hours of playtime telemetry across 4,154 titles from Nintendo, Xbox, and Steam for 1,838 UK and US adults, linked to structured genre metadata and to 6,680 biweekly mental wellbeing assessments (SWEMWBS). Bayesian multilevel within-between models evaluated both hypotheses against a preregistered threshold for practical significance. Total playtime and genre-level playtime both showed the same pattern: associations with wellbeing were consistently under the threshold for practical significance. Wave-to-wave fluctuations in total playtime showed no meaningful link to wellbeing, and differences in habitual playtime across participants were similarly negligible. Contrary to our prediction, genre-level playtime did not meaningfully diverge from this pattern: genre heterogeneity was practically negligible under a hierarchical prior and inconclusive but small in the other specifications, including a multi-membership model for multi-genre games, with no evidence that any genre differed meaningfully from the rest. A daily-level exploratory analysis replicated this pattern. Genre differences appear to be the wrong place to look for gaming's wellbeing effects as genre labels are too coarse a proxy for the psychological experiences that plausibly drive them, which directs attention towards finer-grained characterisations of play.

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Introduction

Video game effects research utilising digital trace data often focuses on a single game or a handful of games to understand the psychological processes and outcomes of digital play (Ballou, Vuorre, et al., 2025; Brühlmann et al., 2020; Johannes et al., 2021; Larrieu et al., 2023). While understandable given the limited resources and lack of objective data available to independent researchers, it has left the field guessing whether identified relationships are unique to the studied titles or whether they apply universally across games.

Identifying how different types of games and gameplay affect wellbeing is vital in light of well-established evidence that playtime in general is not a meaningful predictor of wellbeing, using data from both specific games (Larrieu et al., 2023) and platforms (Ballou, Sewall, et al., 2024; Ballou, Vuorre, et al., 2025). This relationship, however, has not yet been tested using comprehensive multi-platform play data.

Multiple mechanisms have been proposed to explain why aggregate playtime is poorly predictive of wellbeing, including displacement of beneficial activities, satisfaction or frustration of basic psychological needs, and selection effects whereby individuals with lower wellbeing play more (Ballou, Hakman, et al., 2025). Most proposed mechanisms operate at the level of game features or player experience rather than at the level of total time spent; playtime is, at best, a moderator (Ballou, Hakman, et al., 2025). In causal terms, playing is a compound exposure that comes in many versions, so an average association across all of them answers an ambiguous question (Magnusson et al., 2024). The few causal analyses to date, exploiting console lotteries in Japan as natural experiments, found positive effects of console ownership on wellbeing that differed by console type and replicated in the post-pandemic period (Egami et al., 2024; Egami et al., 2026), suggesting that observational null findings may reflect confounding rather than absence of effect.

Game Genres and Wellbeing

Consistent null findings for the effects of playtime on wellbeing have generated calls for the unpacking of raw decontextualised playtime into more informative measures that consider games' varied characteristics and affordances (Ballou, Sewall, et al., 2024). Game genres

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This study was preregistered as part of a Stage 1 Programmatic Registered Report; preregistration materials are available at https://osf.io/mvngt/. The Open Play dataset (v1.2.1) is openly available on Zenodo at https://zenodo.org/records/20119134. All analysis code, including the multi-membership model validation simulation, is available at https://github.com/Thomhak/open-play-genres.

offer a structured way to differentiate between video games and their corresponding play experiences (Starosta et al., 2024). Genre differences in player behaviour, cognition, and gaming disorder symptoms are well established (André et al., 2024; Dobrowolski et al., 2015; Hazel et al., 2022; Raith et al., 2021). Action games have been linked to increased visual attention (Palaus et al., 2017), cognitive and attentional control (Anguera et al., 2013; Bavelier & Green, 2019), and working memory (Blacker et al., 2014). Massively multiplayer online role-playing games (MMORPGs), first-person shooters, real-time strategy games (RTS), and Multiplayer Online Battle Arena (MOBA) games have been associated with higher gaming disorder symptoms (Elliott et al., 2012; King et al., 2019; Na et al., 2017; Rehbein et al., 2021). Player experience also differs by genre: MOBA players report less immersion and autonomy and more frustration than players of single-player genres (Johnson et al., 2015), and role-playing games are the genre players most often perceive as beneficial (Hazel et al., 2022). Randomised controlled trials with casual games and exergames induced positive moods and reduced stress (Huang et al., 2017; Russoniello et al., 2009). These findings suggest that the effects of video games are not uniform and that genre differences exist in both gaming behaviour and its impact on wellbeing, however the effects of genre-level playtime on wellbeing are yet to be studied using digital trace data.

With more than 10,000 new games released each year (Notis, 2021), an accessible classification system is essential for studying and comparing video games at scale. Genres, while defined by convention and neither exclusive nor exhaustive as labels (Clarke et al., 2017; Starosta et al., 2024; Vargas-Iglesias, 2020), remain the most pragmatic means of categorising video games, providing utility for researchers, players, industry, and policymakers alike. Psychologists have differentiated games based on genre for decades (Griffiths, 1993; Wolf, 2002), but must negotiate challenges in definition, overlaps between genres, and the continuous evolution of genres as new games are released. Genre labels conflate distinct facets of games, such as gameplay style, purpose, and mood (Lee et al., 2014); treating players as pure-genre versus acknowledging multi-genre play alters conclusions about player characteristics (Azizi et al., 2018), and factor-analytic decompositions of crowd-sourced game tags produce genre structures that diverge from conventional labels (Li, 2020). Even the cognitive literature that produced the best-known genre-specific claims now treats genre labels as a weakening proxy for the mechanics that matter, because games increasingly blend mechanics across genres (Bavelier & Green, 2019). Genres are therefore best treated not as definitive taxonomical statements but as crude yet practical tools for organising a growing and diversifying library.

There is a lack of consistent standardisation of game genres, therefore classification of video game genres across psychological research is highly inconsistent (Starosta et al., 2024). Of 96 examined articles, only 50 (52%) employ an existing genre classification; 46 (48%) use a unique classification, and 18 (19%) fail to specify the basis or definitions of their genre categorisations altogether. Genre classifications are typically self-reported by participants or assigned by authors, and the same games are habitually classified into different genres across studies. This subjectivity undermines the validity of existing self-report or researcher-ascribed genre taxonomies for the meaningful comparison of results across studies.

Structured, crowd-sourced metadata repositories such as SteamDB, The Games Database, and the Internet Games Database (IGDB) offer a more reliable alternative, involving both game developers and large player samples in collaboratively maintaining genre labels (Lee et al., 2015; Li & Zhang, 2020; Starosta et al., 2024). Within the constraints of genre as a means of categorising games, metadata repositories best accommodate the fluidity of genres over time (Cohen, 1986) and enable comparative research at scale.

The Present Study

In line with prior evidence (Ballou, Sewall, et al., 2024; Ballou, Vuorre, et al., 2025; Larrieu et al., 2023), our first hypothesis predicts the absence of a meaningful relationship between playtime and wellbeing, this time with digital trace data across several platforms and thousands of games:

H1. Total cross-platform playtime is not meaningfully associated with fluctuations in general mental wellbeing over a 2-week period (**H1a**, "within-person") or with average wellbeing over the full study period (**H1b**, "between-person").

We test this via a Bayesian ROPE analysis with equivalence bounds of ±0.06 SWEMWBS item-mean points per hour/day, matching the SESOI used by Ballou, Sewall, et al. (2024). In practical terms, H1a asks whether playing more than one's own average in a given two-week wave is associated with a shift in wellbeing, while H1b asks whether individuals who habitually play more report different average wellbeing than those who play less.

Second, following previous literature (André et al., 2024; Dobrowolski et al., 2015; Elliott et al., 2012; Hazel et al., 2022; Raith et al., 2021; Rehbein et al., 2021), we predict that genres vary in their relation to wellbeing, with some potentially positive and others potentially negative, varying in strength. Put more formally:

H2. Genres differ in how playtime relates to fluctuations in general mental wellbeing over a 2-week period (**H2a**, "within-person") and to average wellbeing over the full study period (**H2b**, "between-person").

H2a asks: when a participant plays more than their own average this wave, does it matter what genres that extra playtime is in? H2b asks the between-person complement: among people who play more overall, does it matter whether that habitual extra playtime is concentrated in, for example, RPGs versus Shooters versus other genres? H2 thus tests whether it is justified to generalise video game effects across genres, testing the generalisability that single-title studies must assume (Yarkoni, 2022).

Beyond the registered hypotheses, we report one unregistered extension. The biweekly window aggregates many individual play sessions into a single retrospective judgement, so the registered tests leave open whether associations operating on shorter timescales are hidden by aggregation. As such, we re-examine both hypotheses at the daily level. This extension was specified after the registered analyses were complete and is reported as exploratory throughout.

Methods

Data Source and Participants

The analyses reported here are part of a Stage 1 Registered Report (Ballou, Földes, et al., 2024), and the data collected for this study are part of the Open Play dataset (v1.2.1) (Ballou, Földes, Vuorre, et al., 2025), a longitudinal dataset that combined multi-platform video game digital trace data alongside daily and biweekly psychological measures from adult gamers in the UK and US over a three-month period. The intake surveys were completed between Sep 2024 and Jan 2026. Digital trace data span the following periods: Nintendo (May 2022 to Oct 2025), Xbox (May 2022 to Sep 2025), Steam (Nov 2024 to Oct 2025), iOS (Mar 2025 to Aug 2025), Android (Mar 2025 to Aug 2025).

This study received ethical approval from the Social Sciences and Humanities Inter-Divisional Research Ethics Committee at the University of Oxford (OII_CIA_23_107). All participants provided informed consent at the start of the study, including consent to their data being shared openly for reanalysis.

Participants were paid at an average rate of £12/hour, equating to £0.20 for a 1-minute screening, £2 for the 10-minute intake survey (plus £5 for linking at least one account with recent data), £0.80 for each 4-minute daily survey, and £2 for each 10-minute biweekly survey. Participants received a £10 bonus payment for completing at least 24 out of 30 daily surveys and/or 5 out of 6 biweekly surveys.

From 35,369 participants who completed the intake survey, 2,544 met the preregistered screening and linking criteria. The analytic sample comprises the 1,838 participants who provided valid SWEMWBS responses in at least one biweekly survey wave. Playtime for these participants is measured from their linked Nintendo, Xbox, and Steam accounts; participants and waves without recorded play on these platforms are retained with zero hours rather than excluded, and 1,541 participants (84%) had at least one gameplay session of a game that could be matched to a genre. The demographic sensitivity analysis uses the 1,741 participants with a matching qualified intake record and age within the 18-40 screening range (1,041 US, 700 UK; mean age 26.9 years, 66% men); 97 analytic-sample participants are excluded from that subset (97 without a matching intake record, 0 aged over 40). Full demographic characteristics, including country-level breakdowns, are reported in Table A0.1.

Measures

The distributions and structure of the key variables are visualised in Figure 2.

Wellbeing. Wellbeing was assessed using the Short Warwick-Edinburgh Mental Well-being Scale (SWEMWBS) (Tennant et al., 2007), a 7-item short form of the WEMWBS covering psychological functioning and subjective wellbeing over the past 2 weeks, with responses on a 5-point Likert scale ranging from “None of the time” to “All of the time.” Scores are computed as the arithmetic mean of the 7 items (range: 1–5). Internal consistency across all analytic person-waves was high (McDonald’s ω_t = 0.89, Cronbach’s α = 0.89). All reported coefficients, priors, and the preregistered SESOI are defined on this item-mean scale.

Video game playtime. Digital trace data from Nintendo, Xbox, and Steam provide session-level records (per-title start and end timestamps), while iOS and Android contribute only daily aggregate minutes (no game-level metadata). Nintendo, Xbox, and Steam therefore contribute to both H1 (total playtime) and H2 (genre-specific playtime); iOS and Android contribute only to the preregistered all-platform sensitivity analysis (Appendix A, Deviation D4).

Game genre classification. Game genre labels for Nintendo and Steam are obtained by cross-referencing game titles in the digital trace data with the Internet Games Database (IGDB). IGDB is, to our knowledge, the only database with complete multi-platform metadata coverage that offers an API for programmatic data retrieval. The genre labels are crowd-sourced: community members can submit contributions (e.g., a new game or alternative categorisation), which are vetted by admins and moderators before appearing in the database. For Xbox, game titles could not be matched to IGDB genre labels, since Xbox provided us only with anonymised title identifiers. Instead, genre labels were available through the Xbox store and were mapped onto the IGDB taxonomy. In total, we collected 23 distinct raw IGDB genre labels, converted the Xbox store genres into the IGDB taxonomy, and collapsed them into 14 study-level genre categories (e.g., Strategy subsumes Turn-Based Strategy (TBS), Real-Time Strategy (RTS), Tactical, and MOBA; Puzzle subsumes Card & Board Game and Quiz/Trivia). The complete mapping table can be found in Appendix B, and the descriptive statistics by genre in Appendix C. We did not consider IGDB “themes” or other metadata fields. Overall, 94.1% of in-study playtime is matched to at least one study genre (Steam 96.3%, Nintendo 92.3%, Xbox 80.2%), with unmatched playtime counting towards the total playtime but towards no genre.

Genre co-occurrence and compositional multi-genre attribution. Across the 3,721 individual games in our dataset, 62% carry more than one genre label, leading to 325 distinct genre combinations. Figure 1 shows the playtime-weighted co-occurrence structure across the 40 most-played combinations, and Table 1 lists the ten most-played combinations with their top titles. Operationalising games as multi-genre has consequential effects in relation to playtime and can be handled in one of three ways, which differ in how they construct the genre-exposure variable and how they enter the models.

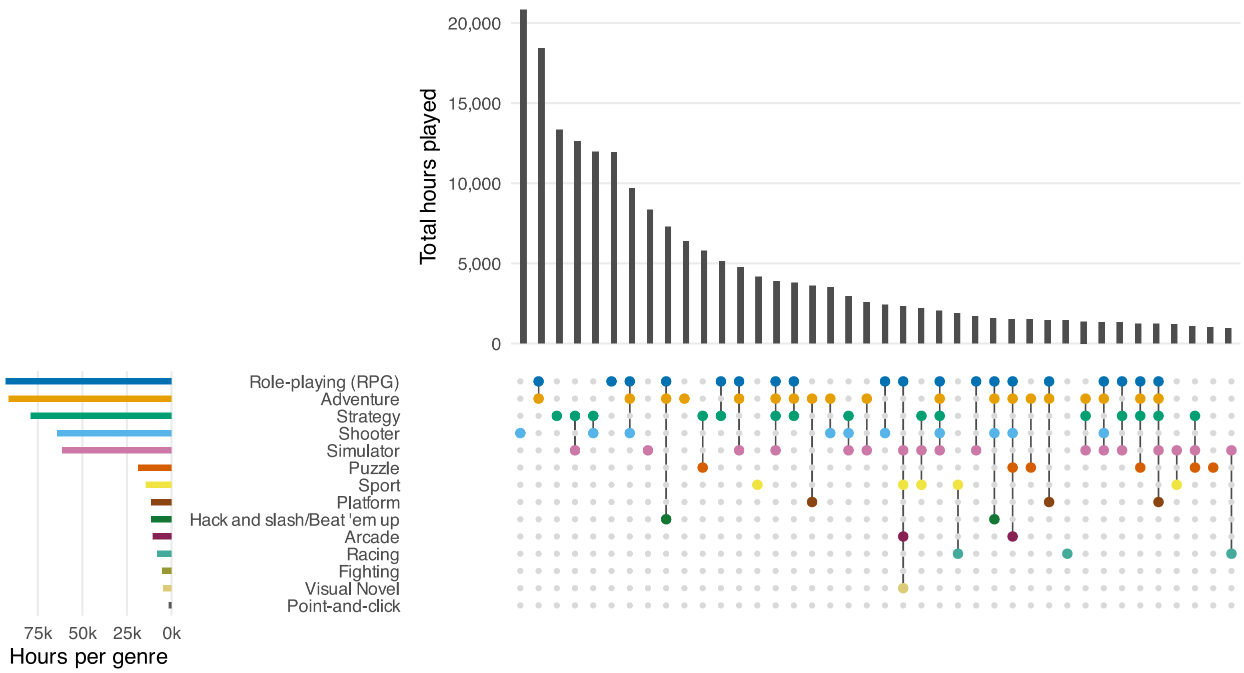

Figure 1. Genre Co-Occurrence Weighted by Total Playtime. Top panel: total hours played by games in each genre combination, ordered by playtime. Bottom panel: dot matrix indicating which genres belong to each combination; connected dots denote multi-genre games. Left bars show total hours attributed to each genre, counting a multi-genre game's full playtime towards each genre exposure.

Partial attribution splits a session's playtime across its genres. A 60-minute session of *Skyrim* (Adventure and RPG) would be allocated as 30 minutes of Adventure and 30 minutes of RPG (or potentially a different, unequal partition). This preserves the constraint that exposure sums to total playtime but penalises multi-genre games, since the more genres a game carries, the less exposure it contributes to each, even though the game belongs fully to each genre, not partially.

Full attribution counts a session's full duration towards every genre it carries. The same 60-minute *Skyrim* session counts as 60 minutes of Adventure and 60 minutes of RPG. Consider a person who plays 60 minutes of *Skyrim* (Adventure + RPG) and 60 minutes of *Final Fantasy XIV Online* (RPG only). Under partial attribution, *Skyrim* contributes 30 minutes to each genre, yielding 30 total minutes of Adventure and 90 minutes of RPG, a 1:3 ratio implying three times as much RPG engagement as Adventure. Under full attribution, *Skyrim* contributes 60 minutes to each genre, yielding 60 minutes of Adventure and 120 minutes of RPG, a 2:1 ratio that better reflects that RPG-labelled content received twice the engagement of Adventure-labelled content. A consequence is that genre exposures no longer sum to total playtime. Genre-level coefficients therefore represent the association with an added hour of that genre's exposure holding total playtime constant, for example, replacing a single-genre RPG session with a dual-genre Adventure/RPG session of the same length.

Proportional multi-membership weighting converts full attribution hours to compositional weights by dividing each genre's exposure hours by total exposure (e.g., 33.3% Adventure, 66.7% RPG for a wave consisting of 60 minutes of *Skyrim* and 60 minutes of *Final Fantasy XIV Online*) and uses these as multi-membership weights on a single total-playtime slope. Genre does not enter the model (M4) as an independent predictor of wellbeing but as a modifier of the playtime–wellbeing slope. Thus, a wave consisting mostly of RPG play has its playtime coefficient adjusted mostly by the RPG slope modifier, with other genres contributing in proportion to their share of that wave's play. This weighting shares a limitation with partial attribution: adding a genre label to a game lowers the shares of the labels it already carries, so none of the three options is free of trade-offs. The genre-specific slope adjustments are drawn from a common distribution whose standard deviation, σ_s, is M4's answer to H2.

Demographics. Age, gender, ethnicity, country, and employment were collected at baseline via self-report. Full demographic characteristics are reported in Table A0.1 in Appendix C.

Analytic Approach

All analyses were conducted in R (R Core Team, 2025), and the manuscript was written in Quarto (Allaire et al., 2025). We fit four Bayesian multilevel models with `brms` (Bürkner, 2017) at the biweekly timescale, and refit each at the daily timescale as an exploratory extension. Two were preregistered: the random-effects within-between (REWB) regression of total playtime (M1) tests H1, and the genre model with independently estimated

Genre combination	Total hours	% of playtime	n games	Mean min/day	Top 5 titles
Shooter	20820	10	268	13	Marvel Rivals [37.2%], Overwatch 2 [11.6%], Call of Duty: WWII [6%], Team Fortress 2 [3.9%], PUBG: Battlegrounds [3.8%]
Adventure + Role-playing	18425	8	213	12	The Elder Scrolls V: Skyrim - Special Edition [8.3%], Clair Obscur: Expedition 33 [6.6%], The Elder Scrolls IV: Oblivion Remastered [5.9%], Elden Ring [5.4%], Throne and Liberty [4.5%]
Strategy	13352	6	139	8	Dead by Daylight [30.9%], R.E.P.O. [16.7%], Dota 2 [11.7%], DeadLock [6.6%], Bloons TD 6 [6.4%]
Simulator + Strategy	12621	6	204	8	RimWorld [18.5%], Schedule I [17%], Europa Universalis IV [6.3%], Hearts of Iron IV [6.3%], Factorio [5.8%]
Shooter + Strategy	11952	5	34	8	Counter-Strike 2 [41.6%], Helldivers 2 [17.5%], Apex Legends [12.7%], 5268BC7F-B83E-4277-9B5A-EF6C2A514F81 [9.5%], Tom Clancy’s Rainbow Six Siege X [6.9%]
Role-playing	11942	5	138	8	Final Fantasy XIV Online [24.7%], Elden Ring: Nightreign [22.4%], Lightning Returns: Final Fantasy XIII [6.1%], Star Wars: The Old Republic [4.6%], Xenoblade Chronicles X: Definitive Edition [3.1%]
Adventure + Role-playing + Shooter	9689	4	50	6	Warframe [24.5%], Destiny 2 [15.1%], Cyberpunk 2077 [10.3%], Red Dead Redemption 2 [9%], Rust [7.6%]
Simulator	8337	4	212	5	7DAC9E96-85E7-479B-9622-8A7E73C23FA2 [15.8%], Animal Crossing: New Horizons [11%], Garry’s Mod [8.8%], CAC402B8-C0BC-4A0D-AFD3-95E5BAD45B34 [6.6%], House Flipper 2 [4.9%]
Adventure + Hack and slash/Beat ‘em up + Role-playing	7293	3	52	5	Path of Exile [27.9%], Monster Hunter Wilds [19.8%], Path of Exile 2 [12.1%], Last Epoch [5.4%], Torchlight: Infinite [5.1%]
Adventure	6381	3	358	4	2089FA57-90DD-4B9A-B9C0-43013F5D6E70 [7.9%], C3A5EDE3-F40F-489E-8E4C-EEAA3BD904C9 [5.6%], F4FF1A75-054D-4451-B506-5AE87E111E2C [5%], BE4D7FFB-533D-4185-8E8A-90CEBFF8858D [4.4%], C1E17D99-3740-44FE-B489-0E8B1D728D6E [4.3%]

Table 1. Top 10 genre combinations by total playtime. Multi-genre rows join labels with ‘ + ‘. Total hours = playtime for games with that combination; % of playtime = share of all genre-matched playtime; n games = unique game-platform pairs. Xbox titles are anonymised.

genre coefficients (M2) tests H2. Two are exploratory and were added after the M2 results were known: M3 keeps M2′s predictors but pools the genre coefficients through a common prior, whereas M4 keeps total playtime as the single predictor, as in M1, and lets each wave’s genre composition modify its slope; both answer H2 through the estimated spread of genre effects, and M4 also yields H1 estimates. The Results are organised by hypothesis; within each, the preregistered model comes first and the exploratory models second, and every figure row, legend entry, and table row is labelled by model with its registration status. Deviations from the preregistered plan are catalogued in Appendix A and are flagged below where they affect interpretation.

Smallest Effect Size of Interest

Establishing the absence of a practically significant effect requires committing in advance to a threshold below which an association is too small to matter, the smallest effect size of interest (SESOI). By practically significant, we mean the smallest change in wellbeing that a player would themselves notice, known also as a minimally important difference (MID). Following previous studies, the MID is set at 0.3 points on the 1–5 SWEMWBS scale, converging with anchor-based estimates of the Warwick-Edinburgh Mental Well-being Scale (WEMWBS) (Ballou, Sewall, et al., 2024; Maheswaran et al., 2012) and the Positive and Negative Affect Schedule (PANAS) (Anvari & Lakens, 2021).

To calculate the SESOI, we also need to know how large a change in playtime is realistically available to reach the MID. Because UK and US adults average approximately 5 hours of daily leisure time (Sturm & Cohen, 2019), this provides us with an upper bound on how much daily play could plausibly rise or fall without it displacing other behaviours that arguably impact wellbeing more than play (e.g., sleep). SESOI is the ratio of the two: 0.3 ÷ 5 = 0.06 points per hour. An association smaller than this means that even a person who shifted their entire 5-hour daily leisure time budget into or out of gaming would not experience a noticeable change in wellbeing.

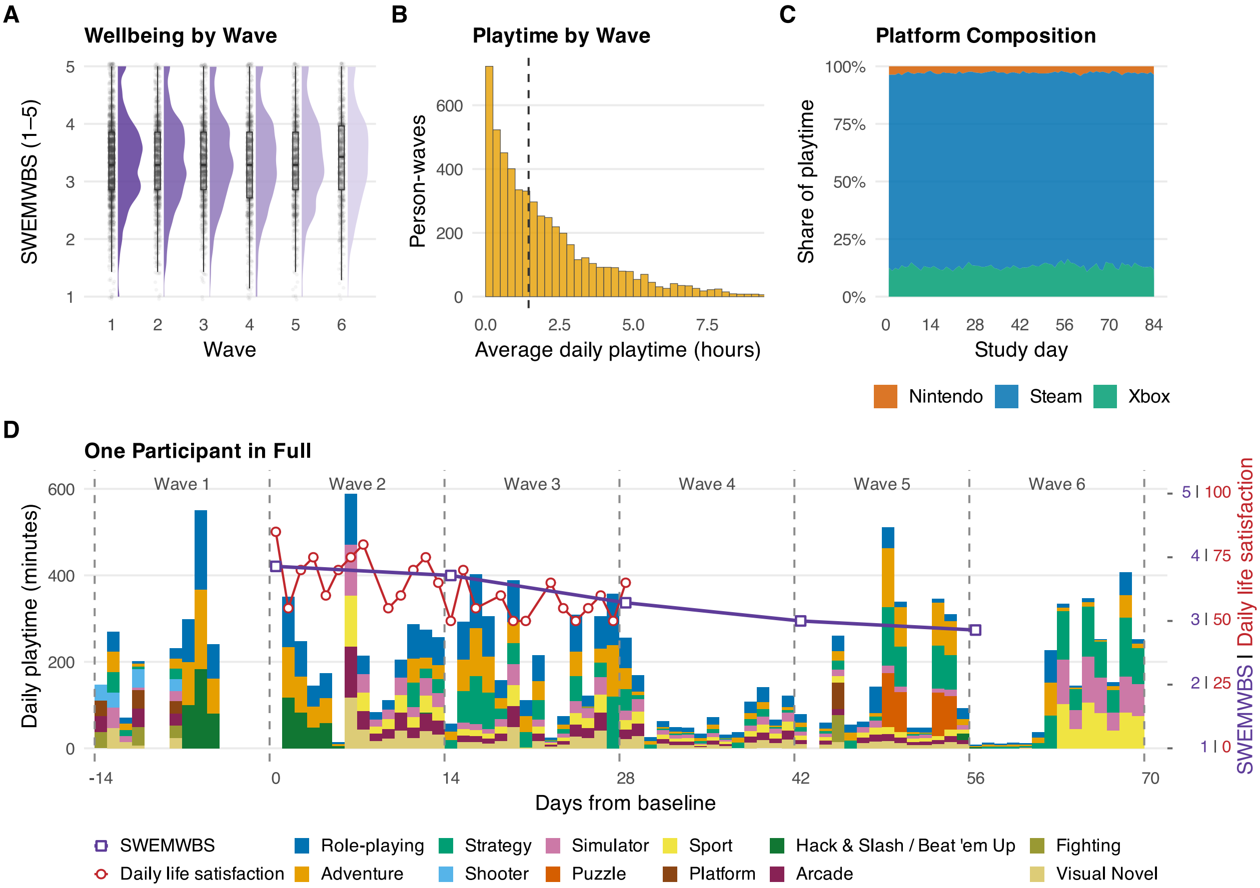

Figure 2. Distributions, Composition, and Raw Structure of the Study Measures. (A) SWEMWBS distributions per biweekly wave (density, raw points, and median box). (B) Average daily hours per person-wave with recorded play (15-minute bins; dashed line: median). (C) Share of recorded playtime per study day by platform, days 1–84. (D) Daily playtime for one participant (ID p5271179686); bar height is genre-matched playtime, segments are its genre composition under proportional multi-membership weighting. Dashed lines mark the six 14-day waves. Two outcomes are overlaid on the right axis: biweekly SWEMWBS (purple open squares, 1–5) at each survey date, and daily life satisfaction from the 30-day diary (red open circles, 0–100).

We evaluate all individual coefficients (H1) against the interval between the negative and positive SESOI [−0.06, +0.06], also known as the Region of Practical Equivalence (ROPE) (Kruschke, 2018). If a coefficient's 95% credible interval (CrI) falls entirely inside the ROPE, the effect is considered practically null, meaning even the largest plausible effect is too small to matter; a CrI entirely beyond either bound indicates a practically positive or negative effect, and one straddling either edge is inconclusive.

Evaluating genre heterogeneity (H2) requires a modified rule, because it represents a spread of differences rather than a single effect. We consider genre differences to be practically negligible only if 95% of the genre-specific effects fall inside the ROPE. Because 95% of a normal distribution lies within ±1.96 standard deviations of the centre, fitting this distribution inside our 0.06 ROPE requires the spread (standard deviation) to be below 0.031 (0.06 ÷ 1.96). If the posterior probability of meeting this condition is less than 95%, the heterogeneity is considered inconclusive. To make this spread easier to interpret, we also report the expected number of our 14 genres whose effects are large enough to fall beyond the SESOI.

Preregistered Models

M1: Total Playtime

M1 is the preregistered model for H1: is total playtime across Nintendo, Xbox, and Steam associated with wellbeing? The model is a REWB regression and returns the two coefficients, β_W for within-person fluctuations and β_B for habitual between-person differences. Each participant receives a random intercept, because people differ in baseline wellbeing, and a random slope for within-person playtime, because the association need not be identical for everyone. Priors are calibrated to the 1–5 outcome scale. The intercept prior is centred on the scale midpoint, and the coefficient priors are weakly informative, leaving effects up to roughly 1 point per hour/day plausible while regularising implausibly large values.

$$
\begin{aligned}
\text{SWEMWBS}_{ij} &\sim \text{Normal}\ (\mu_{ij}, \sigma) \\
\mu_{ij} &= \alpha + \beta_W \text{Time}_{\text{within},ij} + \beta_B \text{Time}_{\text{between},j} + u_{0j} + u_{1j} \text{Time}_{\text{within},ij} \\
\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} &\sim \text{Normal}\ (\mathbf{0}, \mathbf{\Sigma}) \\
\alpha &\sim \text{Student-}\ t(3, 3, 2) \\
\beta_W, \beta_B &\sim \text{Student-}\ t(3, 0, 0.5)
\end{aligned}
\tag{1}
$$

In this and the following model equations, parameters that are not listed take the `brms` default priors: the residual standard deviation σ and the standard deviations of the participant-level effects have half-Student-t(3, 0, 2.5) priors, and correlation matrices have LKJ(1) priors, which are uniform over the correlation in the two-dimensional case.

M2: Independent Genre Coefficients

M2 is the preregistered specification for H2. Each of the 14 genres receives its own within-person and between-person coefficient, estimated independently, with a random intercept per participant. Because genre exposures are non-exclusive, they cannot be summed to recover total playtime, and M2 therefore contains no separate total-playtime term. Two departures from the registered specification should be noted: (1) exposures use full attribution rather than primary-genre labels (Deviation D1), and (2) the registered random slopes for all genre terms (28 additional parameters) proved near-singular, so we followed the preregistered contingency plan and retained the random intercept only.

$$
\begin{aligned}
\text{SWEMWBS}_{ij} &\sim \text{Normal}\ (\mu_{ij}, \sigma) \\
\mu_{ij} &= \alpha + \sum_{g=1}^{G} (\ \beta_{W,g}\ \text{Time}_{\text{within},g,ij} + \beta_{B,g}\ \text{Time}_{\text{between},g,j}\ ) + u_{0j} \\
u_{0j} &\sim \text{Normal}\ (0, \tau_{\text{pid}}) \\
\alpha &\sim \text{Student-}\ t(3, 3, 2) \\
\beta_{W,g}, \beta_{B,g} &\sim \text{Student-}\ t(3, 0, 0.5)
\end{aligned}
\tag{2}
$$

Exploratory Models

The models in this section were not preregistered and are reported as exploratory throughout, together with the daily-timescale refits described at the end of this section.

M3: Hierarchical Prior

M3 keeps M2′s structure but changes how the genre coefficients are estimated, and this change yields a direct answer to H2. Rather than treating the 14 coefficients as unrelated, M3 assumes that they are drawn from a common normal distribution centred at zero and estimates the standard deviation of that distribution, τ_{genre}, separately for the within-person and between-person dimensions. τ_{genre} is therefore an estimate of exactly the quantity H2 asks about: how much the genre coefficients differ from one another, expressed in outcome units per hour of play. The shared distribution also produces partial pooling. Estimates for rarely played genres are pulled towards the common centre, whereas genres with ample data retain estimates close to their independent values.

The Half-Normal(0, 0.1) hyperprior on τ is matched to the outcome scale. Genre coefficients are expressed in SWEMWBS points per hour/day, and the smallest effect size of interest is 0.06 on that scale, so a much wider hyperprior would place substantial prior mass on heterogeneity far larger than any effect considered plausible. The sensitivity of the results to halving and doubling this scale is examined in Appendix D. Sampling near τ = 0 is numerically difficult, so we use conservative sampler settings and report divergence counts and R-hat diagnostics for every model alongside the results.

$$
\begin{aligned}
\text{SWEMWBS}_{ij} &\sim \text{Normal}\ (\mu_{ij}, \sigma) \\
\mu_{ij} &= \alpha + \sum_{g=1}^{G} (\ \beta_{W,g}\ \text{Time}_{\text{within},g,ij} + \beta_{B,g}\ \text{Time}_{\text{between},g,j}\ ) + u_{0j} \\
u_{0j} &\sim \text{Normal}\ (0, \tau_{\text{pid}}) \\
\beta_{W,g} &\sim \text{Normal}\ (0, \tau_{\text{within}}) \\
\beta_{B,g} &\sim \text{Normal}\ (0, \tau_{\text{between}}) \\
\tau_{\text{within}}, \tau_{\text{between}} &\sim \text{Half-Normal}\ (0, 0.1) \\
\alpha &\sim \text{Student-}\ t(3, 3, 2)
\end{aligned}
\tag{3}
$$

M4: Multi-Membership

M4 represents each wave the way panel D of Figure 2 displays each day, namely as a total amount of playtime with a genre composition. The model retains total playtime as the predictor, inheriting M1′s structure including the overall β_W and β_B, and allows the composition of that playtime to modify the playtime slope. Formally, this is a multi-membership structure, where each wave carries all 14 genre labels and only the weights vary, equal to each genre's share of exposure. Waves with no recorded play receive equal weights of 1/14. As such, a wave consisting mostly of Shooter play is governed mostly by the Shooter slope adjustment, and a wave divided between Adventure and RPG draws on both adjustments in proportion.

Each genre's slope adjustment is drawn from a common distribution, and the standard deviation of that distribution, σ_s, is this model's answer to H2: it quantifies how much the playtime–wellbeing slope varies with the genres being played. A genre's within-person and between-person adjustments are drawn jointly, with a correlation between them estimated from the data.

$$
\begin{aligned}
\text{SWEMWBS}_{ij} &\sim \text{Normal}\ (\mu_{ij}, \sigma) \\
\mu_{ij} &= \alpha + \beta_W \text{Time}_{\text{within},ij} + \beta_B \text{Time}_{\text{between},j} + u_{0j} + u_{1j} \text{Time}_{\text{within},ij} \\
&\quad + \sum_{g=1}^{G} w_{g,ij} (s_{W,g} \text{Time}_{\text{within},ij} + s_{B,g} \text{Time}_{\text{between},j}) \\
\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} &\sim \text{Normal}\ (\mathbf{0}, \mathbf{\Sigma}_u) \\
\begin{pmatrix} s_{W,g} \\ s_{B,g} \end{pmatrix} &\sim \text{Normal}\ (\mathbf{0}, \mathbf{\Sigma}_s), \quad \mathbf{\Sigma}_s = \begin{pmatrix} \sigma_s^2 & \rho\, \sigma_s \sigma_{s,\,\text{between}} \\ \rho\, \sigma_s \sigma_{s,\,\text{between}} & \sigma_{s,\,\text{between}}^2 \end{pmatrix} \\
\alpha &\sim \text{Student-}\ t(3, 3, 2) \\
\beta_W, \beta_B &\sim \text{Student-}\ t(3, 0, 0.5) \\
\sigma_s, \sigma_{s,\,\text{between}} &\sim \text{Half-Normal}\ (0, 0.3) \\
\rho &\sim \text{Uniform}\ (-1, 1)
\end{aligned}
\tag{4}
$$

Because this use of multi-membership is unconventional, we validated the specification by parameter recovery before fitting it to real data. Synthetic datasets with known genre-slope heterogeneity were fit with identical model code; every population-level parameter, including σ_s, was recovered within its 95% credible interval, with no divergent transitions (simulation report in the supplementary materials).

Daily Timescale

To assess whether our findings hold on a shorter timescale, we conducted an exploratory daily-timescale analysis. US participants were asked to complete daily surveys for the first 30 study days, including a life-satisfaction item ("I was satisfied with my life today") answered on a 0–100 slider (Ballou, Földes, Vuorre, et al., 2025), which serves as the outcome. The exposure is console and PC playtime from 4:00 a.m. local time of the response's waking day up to the survey completion timestamp, computed from the same session-level telemetry. Out of 15.8K diary responses from 1.2K participants, 13.0K responses (82.6%) from 763 participants had available telemetry and analysable same-day windows and were included in the analysis. Playtime is again split into a within-person and a between-person part, but the two parts are measured over different windows. The within-person part is the difference between the day's same-day playtime and the participant's own average same-day playtime. The between-person part is the participant's habitual playtime over a full day, so that differences between participants remain expressed in the same daily-hour units as in the biweekly analysis. The SESOI for the slider is constructed in the same way as the registered one, as a minimally important difference divided by the roughly five hours of daily leisure realistically available (Sturm & Cohen, 2019). No anchor-based minimally important difference exists for a single-item life-satisfaction slider, so we set it at 5 slider points, a deliberately conservative choice. It lies below two available benchmarks, half the within-person SD of the slider (6.3 points) and the registered minimally important difference expressed as a share of its scale range (7.5% of 0–100, or 7.5 points), and a smaller minimally important difference makes practical equivalence harder to claim rather than easier. Dividing by five hours gives 1 slider point per hour of daily play. The model specifications (M1–M4) are refit at the daily level and are labelled M1-D to M4-D. Window construction, sample exclusions, and four robustness refits are detailed in Appendix D.

Missing Data

Missingness in the outcome takes two forms. Item-level missingness is rare and all-or-nothing, with 43 submitted surveys missing all seven SWEMWBS items. Wave nonresponse is the dominant form of missingness, survey completion declined from 100% at wave 1 to 40% at wave 6, leaving 4,348 of the 11,028 person-waves (39.4%) without an observed SWEMWBS.

The Bayesian models are fit to the observed outcomes; this is valid provided outcome missingness is at random given the modelled covariates and random effects. To test that assumption, we multiply imputed SWEMWBS on the complete participant × wave grid with two-level multiple imputation by chained equations (`mice` and `miceadds`) (Robitzsch & Grund, 2024; van Buuren & Groothuis-Oudshoorn, 2011): predictive mean matching (`2l.pmm`) with participants as random-intercept clusters (van Buuren, 2018), and m = 100 imputations, set from the pilot fraction of missing information by von Hippel's (2020) two-stage rule. Predictive mean matching donates observed values, so imputed scores respect the 1–5 scale. Frequentist counterparts of the analysis models were fit to each imputed dataset and pooled by Rubin's rules. The pooled results do not meaningfully differ from the complete-case analyses (Appendix D, Table A0.4). Playtime was not imputed because tracking was continuous and automatic even for participants with missing survey waves, so within linked platforms an absence of play is recorded absence rather than missing data, and exposures exist for every person-wave.

Results

The analytic sample comprised 1,838 participants with a valid SWEMWBS response in at least one biweekly wave, yielding 6,680 person-wave observations (M = 3.6 of 6 waves per participant; see Missing Data). Playtime is measured from linked Nintendo, Xbox, and Steam telemetry; participants played an average of 1.67 hours per day (SD = 2) across the three platforms.

We report the preregistered data quality controls first and then each hypothesis in turn. Within each hypothesis section, the preregistered model (M1 for H1, M2 for H2) comes first and the exploratory models (M4 for H1; M3 and M4 for H2) second, and figure rows, legend entries, and table rows are labelled by model and registration status. The exploratory daily-timescale refits and the robustness checks close the section.

Data Quality Controls

Before testing the hypotheses, we ran the three preregistered positive controls, each of which checks that the data reproduce a pattern known to hold.

First, self-reported biweekly playtime correlated positively with digital trace playtime (r = 0.469, 95% CI [0.447, 0.49], p < .001).

Second, session integrity largely held. Xbox contained 0 overlapping sessions, and no Steam session's recorded playtime exceeded its elapsed wall-clock duration (0 violations). Nintendo retained a small number of overlapping sessions: 675 of 288,082 sessions (0.23%), with a median overlap of 9.8 minutes and accounting for 0.09% of total Nintendo playtime.

All overlapping sessions could be traced to concurrent use of multiple consoles linked to one account (e.g., a Switch Lite overlapping an OLED model). Given their rarity, we judged them neither a sign of systematic data quality problems nor frequent enough to affect the estimates.

Third, men played more shooter games than women (M = 0.487 vs. 0.309 hrs/day, t(1144.4) = 4.25, p < .001, Cohen's d = 0.21), meeting the preregistered criterion of a significant difference in the expected direction and consistent with population studies of gendered genre preferences (Rehbein et al., 2016). All three controls passed, and we proceeded to the hypothesis tests.

Total Playtime (H1)

H1 predicted that total playtime is not meaningfully associated with wellbeing, whether a person plays more or less than usual in a given wave (H1a) or habitually plays more or less than other people (H1b). The preregistered model is M1, the within-between regression of total playtime; each of its two coefficients is judged against the ROPE of ±0.06 points per hour/day. M1 was estimated with Bayesian methods rather than the registered frequentist equivalence test (Deviation D3). The preregistered frequentist test, reported in Appendix D, reaches the same verdicts. Figure 3 shows the posterior distributions, and Appendix E (Table A0.1) tabulates all hypothesis tests.

Preregistered Model (M1)

Within persons (H1a), the posterior mean was 0.003 SWEMWBS item-mean points per hour/day (95% CrI: [−0.008, 0.014]), with 100% of the posterior inside the ROPE. In waves in which a participant played more than their own average, wellbeing did not differ by a practically meaningful amount.

Between persons (H1b), the posterior mean was −0.017 (95% CrI: [−0.035, 0.001]), with 100% of the posterior inside the ROPE. The estimate leans negative, with a credible interval that reaches zero, but its magnitude is far below the SESOI: a 2 hour/day difference in habitual playtime corresponds to −0.034 points on the 1–5 scale, about 0.8% of the 4-point range.

Both components of H1 are supported under the preregistered decision rule.

M1′s random slope estimates how much the within-person association varies between participants. The SD of the person-specific associations was 0.035 (95% CrI: [0.005, 0.062]), and 3.8% of its posterior lies above the SESOI. Under the model's normal random-slope assumption, this spread combined with the average association implies that the within-person association exceeds the SESOI, in either direction, for about 12% of participants. The Discussion returns to this minority.

Exploratory Model (M4)

M4 keeps total playtime as its predictor and lets the genre composition of that playtime modify the slope, so it also estimates the two H1 coefficients. Its estimates match M1′s. Within persons, β_W = 0.007 (95% CrI: [−0.026, 0.043]), with 99.3% of the posterior inside the ROPE; between persons, β_B = −0.014 (95% CrI: [−0.035, 0.01]), with 100% inside (Figure 3). Allowing the association to depend on which genres were played leaves the total-playtime association unchanged. M2 and M3 have no total-playtime term and do not bear on H1.

Genre Differences (H2)

H2 predicted that genres differ in how playtime relates to wellbeing, within persons (H2a) and between persons (H2b). The preregistered model is M2, which estimates a within-person and a between-person coefficient for each of the 14 genres. The exploratory models M3 and M4 estimate the spread of the genre effects as a model parameter, which separates differences between genres from estimation error. Figure 4 and Figure 5 compare the three models.

Preregistered Model (M2)

The preregistered inference for H2 is a joint Wald test of whether the 14 genre coefficients are equal. We fitted a frequentist version of M2 by maximum likelihood, with the same fixed effects and random intercept, and tested the equality of its coefficients. The null hypothesis sets the 13 differences between the first genre's coefficient and each of the others to zero, which holds only if all 14 coefficients are equal; rejecting it means that at least one difference is non-zero, and hence that at least two genres differ. Within persons, the test rejected equality ($\chi^2(13)$ = 24.9, p = 0.024): at least two genres differ in how a change in their playtime relates to a change in wellbeing. Between persons, the test did not reject equality ($\chi^2(13)$ = 13.8, p = 0.385). Under the preregistered criteria, H2a is supported and H2b is not.

The joint Wald test shows whether the coefficients differ, but not which ones differ or by how much. To estimate the size of the differences from the same model,

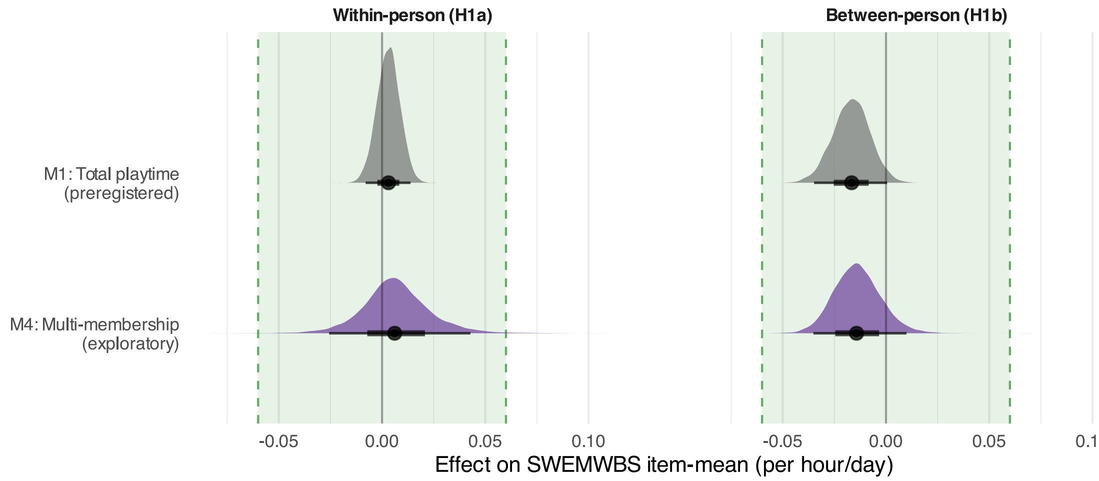

Figure 3. Posterior Distributions for Total Playtime Effects (H1). M1 (grey) is the preregistered total-playtime model; M4 (purple) is the exploratory multi-membership model, which keeps total playtime as its predictor and lets genre composition modify the slope. M2 and M3 have no separate total-playtime predictor and are excluded. Green band: ROPE (±0.06). Thick: 66% CrI; thin: 95% CrI.

we computed the SD of the 14 genre coefficients at each posterior draw of the Bayesian fit of M2 and compared it with the heterogeneity threshold of 0.031, the spread at which 95% of genre effects lie inside the ROPE. This statistic and its threshold were not preregistered. Within persons, the SD averaged 0.032 (95% CrI: [0.02, 0.047]); the interval lies below the SESOI of 0.06 but crosses the threshold, with 43.4% of the posterior below it (Figure 4, blue, top row). Between persons, the SD averaged 0.076 (95% CrI: [0.044, 0.123]), above both thresholds. Because the 14 coefficients are estimated independently, their SD includes each coefficient's estimation error, which is large for rarely played genres. The statistic therefore overstates genre heterogeneity and is an upper limit.

Figure 5 (blue circles) shows the per-genre estimates from M2. The 14 within-person point estimates cluster near zero, and 8 of the 14 credible intervals fall entirely inside the ROPE; the remainder straddle a boundary rather than exclude it. The cost of independent estimation shows in the interval widths: the widest within-person interval (Point-and-Click) is about 5 times wider than the narrowest, too wide to support a conclusion about any rarely played genre. Between persons the intervals are wider still, with 2 of the 14 entirely inside the ROPE.

Exploratory Models (M3 and M4)

M3 and M4 were specified after the M2 results were known. Under M3, the hierarchical prior model, genre variation in the within-person associations (H2a) is practically negligible: τ_{within} = 0.014 (95% CrI: [0.004, 0.028]), with 98.8% of the posterior below the heterogeneity threshold of 0.031 (Figure 4, orange). This spread implies that 0.3% of genres (95% CrI [0%, 3.2%]) deviate from the common centre by more than the SESOI, an expected 0.04 of the 14. Between persons (H2b) the result is inconclusive: τ_{between} = 0.021 (95% CrI: [0.004, 0.05]), with 82.4% of the posterior below the threshold and an implied 2.9% of genres (95% CrI [0%, 22.7%]) beyond the SESOI, an expected 0.4 of the 14.

Under M4, the multi-membership model, the within-person estimate (H2a) is inconclusive: $\sigma_{s,\ \text{within}}$ = 0.044 (95% CrI: [0.006, 0.1]), with 30.4% of the posterior below the heterogeneity threshold (Figure 4, purple). This spread implies that 18.1% of genres (95% CrI [0%, 54.7%]) deviate from the overall playtime slope by more than the SESOI, an expected 2.5 of the 14, and heterogeneity of up to about 0.1 points per hour/day cannot be excluded. The between-person estimate (H2b) is also inconclusive: $\sigma_{s,\ \text{between}}$ = 0.017 (95% CrI: [0.001, 0.046]), with 88.2% of the posterior below the threshold and an implied 2% of genres (95% CrI [0%, 19%]) beyond the SESOI. The posterior is concentrated near zero but does not meet the 95% criterion.

The three estimates in Figure 4 measure the spread of the genre effects in different ways. M2's SD includes estimation error and is an upper limit. M3's τ excludes it and places within-person heterogeneity below the threshold. M4's σ_s also excludes it but is less precise, because it describes how the playtime slope changes with the genre composition of a wave rather than how separate genre exposures differ, and its posterior is too wide to decide.

M3 shrinks the per-genre estimates towards their common centre (Figure 5, orange squares): 14 of the 14 within-person credible intervals fall entirely inside the ROPE, and the wider between-person intervals show the same pattern.

M4 estimates each genre as a deviation from the overall playtime slope (Figure 5, purple diamonds). With the overall fixed effect added back, its within-person estimates

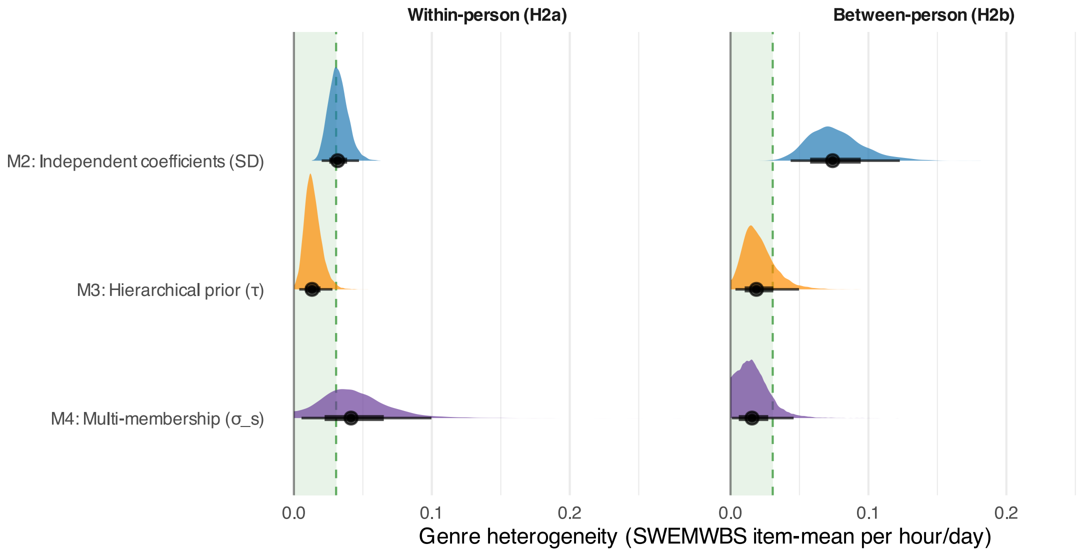

Figure 4. Genre Heterogeneity Under Three Model Specifications (H2). Left: within-person (H2a); right: between-person (H2b). M2 (blue) is the preregistered model (SD of its 14 independently estimated coefficients, which includes estimation error); M3 (orange) and M4 (purple) are exploratory (τ and σ_s, model-estimated spreads). Green band: heterogeneity threshold (0 to 0.031, the spread at which 95% of genre effects lie inside the ROPE of ±0.06). A posterior falling entirely inside the band indicates practically negligible genre heterogeneity. Thick: 66% CrI; thin: 95% CrI.

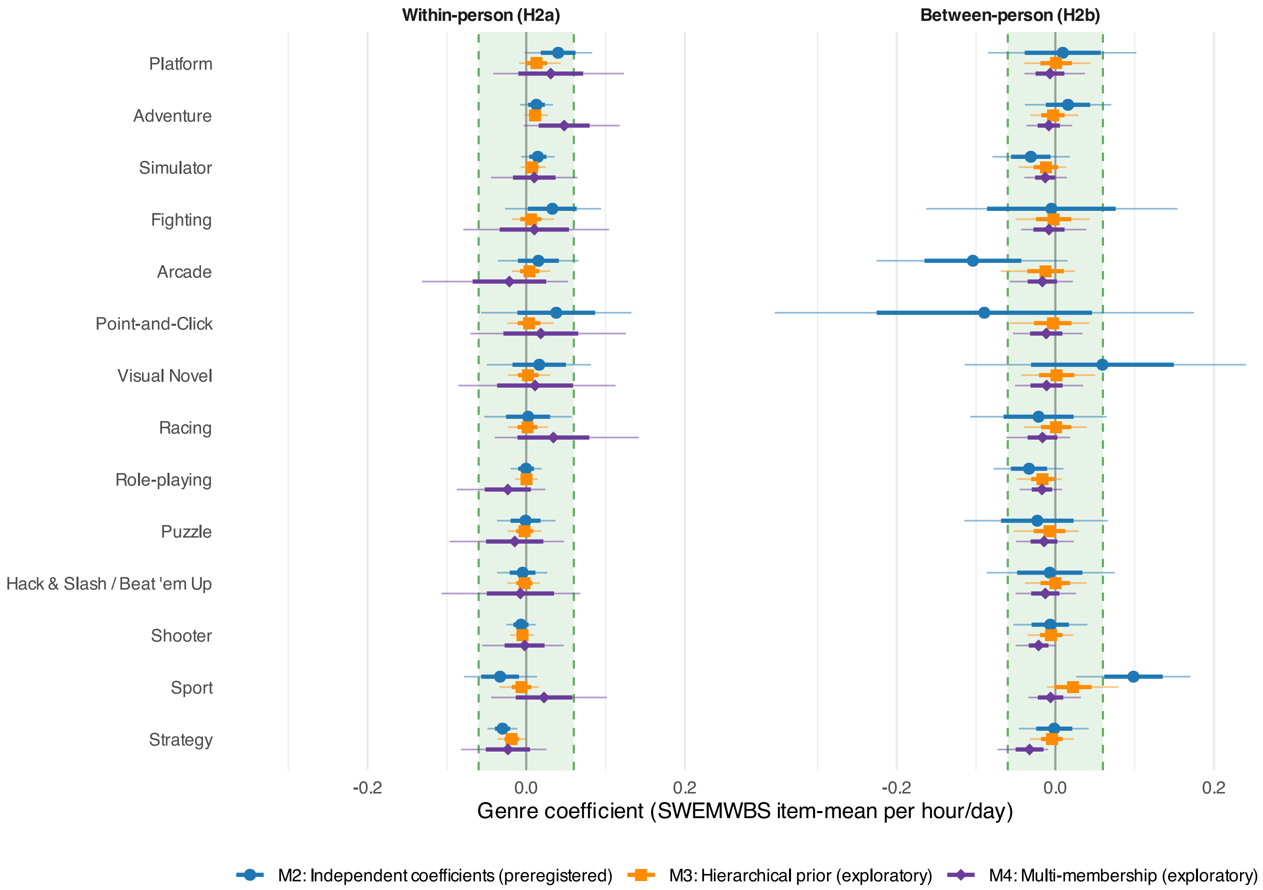

Figure 5. Per-Genre Exposure Coefficients Under Three Model Specifications (H2). Left: within-person (H2a); right: between-person (H2b). Each genre row shows M2 (blue circle; preregistered model, no pooling), M3 (orange square; exploratory, hierarchical prior), and M4 (purple diamond; exploratory, multi-membership). The M2 series is the caterpillar plot specified in the preregistration. Genres ordered by the M3 within-person estimate. Green band: ROPE (±0.06). Thick bars: ±1 posterior SD; thin bars: 95% CrI.

span −0.023 to 0.048 and its between-person estimates −0.033 to −0.006.

In sum, the preregistered test shows that the within-person genre coefficients are not all equal, and no model places the within-person differences at practical significance: they are negligible under M3 and inconclusive but small under M2 and M4. Between persons, M3 and M4 are inconclusive, and M2′s SD, an upper limit inflated by the wide between-person intervals, does not constrain the spread. On neither dimension does any model show a genre that differs meaningfully from the rest. Genre terms do not improve out-of-sample prediction (Appendix D), and the wide intervals for rarely played genres rule out confident inference about any single genre. Appendix E (Table A0.1) tabulates the heterogeneity estimates and convergence diagnostics for the three models; full posterior summaries for all four models are in the supplementary materials.

Daily Timescale (Exploratory)

The daily refits (M1-D to M4-D) were not preregistered, and every estimate in this section is exploratory. Participants in the analysed daily sample contribute a median of 20 diary days, with participation declining from 1,185 responses on day 1 to 411 on day 30; windows close at the survey completion timestamp (median length 13.7 hours), so 60.8% of person-days record no console or PC play.

Figure 6 displays the posteriors against the ROPE. At the daily level, the within-person association between same-day playtime and life satisfaction was positive, at $\beta = 0.18$ points per additional hour in the standalone total-playtime model (M1-D; 95% CrI [0, 0.35]), with the lower credible bound at zero and the entire posterior inside the ROPE (100%); the multi-membership model's fixed effect is similar (M4-D: $\beta = 0.14$, 95% CrI [−0.19, 0.42]). The interval widens past zero under two of the three robustness refits (Appendix D). Person-to-person variation in this slope was practically negligible: the random-slope SD was 0.64 (95% CrI [0.18, 0.97]), with 98.3% of the posterior below the SESOI. Between persons, with habitual play measured in full-day hours, participants who play more than others reported slightly lower daily life satisfaction: M1-D gives $\beta = -0.43$ points per habitual daily hour (95% CrI [−1.11, 0.27]), a negative lean whose posterior sits 95% inside the ROPE; M4-D gives $\beta = -0.46$ (95% CrI [−1.18, 0.27]); and the all-responses refit, which scores

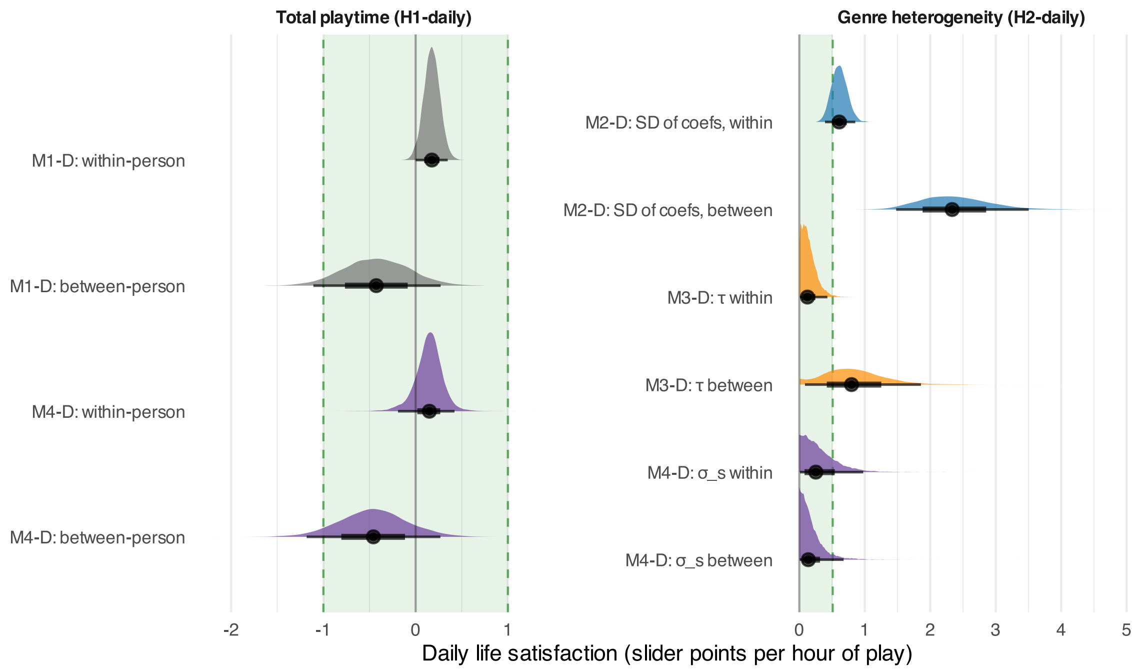

Figure 6. Exploratory Daily-Timescale Posteriors (Life Satisfaction). Left: total-playtime effects from the standalone H1 model (M1-D) and the multi-membership model's fixed effects (M4-D). Right: genre heterogeneity from all three specifications: M2-D (SD across the 14 independently estimated coefficients, computed at each posterior draw), M3-D (hierarchical-prior τ), and M4-D (multi-membership σ_s). Green band: ROPE (±1 slider point per hour) for the total-playtime effects and the heterogeneity threshold (0 to 0.51, 1 ÷ 1.96) for the non-negative spreads. Thick: 66% CrI; thin: 95% CrI.

unobserved play as zero, gives $\beta = -0.52$ (95% CrI [−1.18, 0.15]) (Table A0.2).

Genre heterogeneity at the daily level showed the same pattern as at the biweekly level, judged against the daily heterogeneity threshold of 0.51 slider points (1 ÷ 1.96). Within persons, the hierarchical prior places heterogeneity below the threshold: $\tau_{within} = 0.15$ (95% CrI [0.01, 0.43]), with 99% of its posterior below it and an implied 0.2% of genres beyond the SESOI. The multi-membership model is inconclusive, $\sigma_{s,within} = 0.32$ (95% CrI [0.01, 0.98]; 80.7% below the threshold), as is M2-D, whose SD across the 14 independently estimated genre coefficients of 0.61 (95% CrI [0.39, 0.85]) is computed at each posterior draw and noise-inflated, as at the biweekly level (19.7% below the threshold). The between-person results are inconclusive: $\tau_{between} = 0.84$ (95% CrI [0.09, 1.86]) has 23.9% of its posterior below the threshold; $\sigma_{s,between} = 0.19$ (95% CrI [0.01, 0.68]) has 94%; and M2-D's between-person raw SD (2.37, 95% CrI [1.48, 3.5]) sits far above it, with the usual estimation-error inflation.

Robustness

Three sensitivity analyses test whether the conclusions depend on analytic choices. None changes a conclusion. Adjusting for age, gender, and ethnicity shifts the H1 coefficients by less than 0.01 points and changes no ROPE verdict (Appendix D, Table A0.1). Halving or doubling the heterogeneity hyperprior scales leaves the H2 parameters essentially unchanged (Appendix D, Table A0.2). Refitting on multiply imputed data (m = 100) reproduces the H1 equivalence under the preregistered 90% CI rule (equivalence tests are two one-sided tests at $\alpha = .05$, which correspond to a 90% rather than a 95% interval; Lakens (2017)) and rejects genre-coefficient equality in neither pooled dimension (Appendix D, Table A0.4). Full specifications, results, and diagnostics for all three analyses, together with the predictive model comparison, are reported in Appendix D.

Discussion

When previous work has reported that time spent gaming carries no meaningful association with wellbeing, a common objection is that surely it must depend on the type of game one plays. After all, a relaxing puzzle game and a competitive shooter should not plausibly have the same effect on the person playing them. Video game heterogeneity is the standing explanation for why aggregate playtime nulls should not be taken at face value: playtime is a poorly specified exposure whose associations average over distinct causal contrasts (Ballou, Hakman, et al., 2025; Magnusson et al., 2024), and a null total association can in principle be produced by heterogeneous true effects that offset in aggregate (Yarkoni, 2022). Genre is the most widely used name for those contrasts (Starosta et al., 2024), which makes it the natural first place to look.

Continued debate about how different games might differ in their effects on wellbeing persists in part because of measurement challenges, whereby prior genre research has been subject to at least one of three limitations, and usually multiple. First, genre assignment is inconsistent and unreliable: labels are typically self-assigned by participants or researchers, the same games are classified into different genres across studies, and nearly all taxonomies force each game into a single genre despite modern games being multi-genre by design (King et al., 2019; Starosta et al., 2024). Second, playtime has often been self-reported rather than logged, yet self-reported and logged media use correlate only moderately, with a meta-analytic estimate of $r = 0.38$ and fewer than one in ten self-reports falling within 5% of the logged value (Parry et al., 2021); our own positive control puts the biweekly correlation for playtime specifically at $r = 0.469$, a comparison in which the self-reports may also cover play on platforms outside our telemetry. Third, studies often cover a single game or a handful of titles (Brühlmann et al., 2020; Johannes et al., 2021; Larrieu et al., 2023), leaving genre contrasts to be assembled across different studies that classified and measured both play and genre incommensurably, instead of one study with a single ground truth.

This study clears all three obstacles at once. Genre labels follow a single crowd-sourced taxonomy applied uniformly to every title across all platforms, retaining games' multiple genre labels rather than forcing each into one category. Playtime is measured as objective session-level telemetry across Nintendo, Xbox, and Steam rather than self-reported. And all 14 genres are estimated jointly within a single sample, so genre contrasts emerge from one dataset with a common measurement of play, genre, and wellbeing, rather than being assembled across incommensurable studies.

The playtime-wellbeing association is practically equivalent to zero

When participants played more than their own average in a given two-week wave, their wellbeing barely changed, consistent with previous digital trace studies that have repeatedly found aggregate playtime to be a poor predictor of wellbeing fluctuations (Ballou, Sewall, et al., 2024; Ballou, Vuorre, et al., 2025; Larrieu et al., 2023). Between participants, practical equivalence was likewise established.

Inverting the estimates shows what these associations would amount to even if they were causal. Within persons, wellbeing would shift by one minimally important difference only after roughly 98 additional hours of play per day at the point estimate, and after 22 additional hours at the credible bound most favourable to an association, more extreme than the 10 additional daily hours that Vuorre & Przybylski (2022) derived from game-level telemetry. Between participants, the habitual difference needed to reach the same threshold lies between 9 and 18 hours of daily play. Both requirements exceed the roughly five hours of leisure that adults have in a day (Sturm & Cohen, 2019), however long the difference is sustained, and changes of this order do not occur in practice: of the 1,500 participants observed across adjacent waves, only 121 ever changed their average daily play by 4 or more hours from one wave to the next. An association that no achievable play exposure can reach is, for every practical purpose, indistinguishable from no association at all.

The contrast between light and heavy players deserves one further remark. Participants who habitually play more reported slightly lower average wellbeing, yet the same participants' wellbeing did not fall in waves when they played more than usual. This within-null, between-negative pattern recurs across observational media research and is most plausibly a selection effect, in which people with lower baseline wellbeing spend more time in games rather than games lowering wellbeing (Ballou, Vuorre, et al., 2025; Kievit et al., 2013; Robinson, 1950; Rohrer & Murayama, 2023), although a cumulative effect accruing over years would look the same at these timescales (cf. Vuorre & Przybylski, 2022). Whichever explanation holds, the difference is too small to matter, so public claims about gamers' mental health are, in effect, reports about who plays, not about what playing does.

The causal estimates from Japanese console lotteries point the other way, at 0.2 to 0.5 standard deviations per additional daily hour among lottery compliers (Egami et al., 2024; Egami et al., 2026). Those are local effects of acquiring a new console, measured with self-reported play over the preceding month and diminishing at longer play durations, whereas our within-person estimates describe logged wave-to-wave variation among people who already play. The two estimands need not agree: gaining access to a console and playing an extra hour on a console one already owns are different versions of the same nominal exposure (Magnusson et al., 2024).

Genres do not matter for the playtime-wellbeing association

We expected genres to differ in how playtime relates to fluctuations in general mental wellbeing, however the findings did not support this prediction in any practically meaningful sense. Genre-to-genre variation in the within-person per-hour association was practically negligible under the hierarchical prior and, in the other two specifications, small enough that at most a few of the 14 genres could deviate from the average by more than the SESOI; the 14 within-person genre estimates span only -0.018 to $+0.013$ under the hierarchical prior, with no estimate more than a third of the SESOI from zero, and the most-played genres by total playtime (Role-playing, Adventure, Strategy, Shooter; see Figure 1) sit inside this same narrow band. The preregistered joint test does detect a departure from exact within-person coefficient equality, but of a magnitude well below the SESOI. Genres therefore proved statistically distinguishable but practically interchangeable. The multi-membership model is the least decisive: its fitted spread implies that about 18% of genres (95% CrI [0%, 55%]) deviate from the average slope by more than the SESOI, so heterogeneity confined to a few genres cannot be excluded. Thus, the effect of separate genres cancelling out a general playtime effect is below the threshold of noticeability, and any defence of playtime nulls that appeals to the heterogeneous nature of video games must look to partitions finer than genre.

This result does not contradict the genre differences reported before, because those concern other outcomes and other designs. The most consistent findings involve gaming disorder symptoms, where MMORPG, shooter, and strategy play stand out in cross-sectional surveys of self-reported genre preference (Elliott et al., 2012; King et al., 2019; Na et al., 2017; Rehbein et al., 2021), and player experience, where genres differ in immersion, autonomy, and frustration (Hazel et al., 2022; Johnson et al., 2015). Genre also explains little of why people play: among more than 14,000 highly engaged players, correlations between six gaming motives and genre preference ranged from −0.25 to 0.20 (Király et al., 2022). Our estimand is different, the association between logged exposure to a genre and general wellbeing within and between adults, and on that estimand genres do not separate.

A Susceptible Minority Is Possible but Small

Still, a null average could conceal a minority for whom play genuinely matters (cf. Karhulahti et al., 2025). The random slope in M1 addresses this directly by modelling each participant's own within-person association between playtime and wellbeing. The relevant quantity is not the spread of these slopes but the share of participants the model implies lie beyond the SESOI: combining the posterior for the average within-person effect with the posterior for the person-level slope standard deviation (0.035, 95% CrI [0.005, 0.062]) under the model's normality assumption, an estimated 12% of participants (95% CrI [0%, 33.8%]) have a true within-person association exceeding 0.06 SWEMWBS points per daily hour in either direction. A susceptible minority is therefore compatible with our results rather than ruled out by them; what the data constrain is that such susceptibility is neither common nor, for most of the distribution, large at these timescales. Person-specific slope estimates are heavily shrunk towards the mean given the number of waves per participant, so the model quantifies how heavy the tails are without being able to identify which individuals occupy them; targeted designs with denser within-person sampling would be needed for that.

Shortening the Window Does Not Recover an Effect

A further possibility is temporal, namely that the wellbeing effects of play might surface and dissipate faster than a biweekly retrospective survey can register. To test this claim, the exploratory daily analysis shortens the window from fourteen days to one and reproduces both conclusions. Within persons, every estimate lies inside the equivalence region under every specification, and the same-day point estimate is positive, roughly double its biweekly counterpart in within-person standard-deviation units (0.014 versus 0.009 s_w per hour). The direction is consistent with the session-level affective uplift reported among *PowerWash Simulator* players (Vuorre et al., 2024), though its resolution is fragile across the robustness refits and should be read as suggestive rather than resolved. Its practical magnitude is negligible either way: shifting daily life satisfaction by one minimally important difference (5 points) would require 28 additional hours of same-day play, and within-person genre heterogeneity at the daily level is likewise negligible under the hierarchical prior and inconclusive under the other specifications, as are the between-person counterparts. Therefore, moving the measurement closer in time, on its own, does not conjure a meaningful effect.

Are all games the same?

The question this study set out to answer rests on an undisputed fact, that games differ psychologically, and although current genre labels are not an effective way to operationalise those differences when it comes to wellbeing outcomes, the general intuition persists. Game scholars have argued for decades that genre labels are conventions of marketing and library organisation rather than coherent descriptions of play: they conflate gameplay style, purpose, and mood in a single word (Lee et al., 2014), fail elementary classificatory tests (Clarke et al., 2017), and drift as the medium evolves (Cohen, 1986; Vargas-Iglesias, 2020). Even the literature most committed to genre-specific effects has reached the same conclusion: the researchers who documented attentional benefits of action games now argue that industry genre labels have become poor proxies for the mechanics that produce those benefits, and call for a taxonomy of the processes engaged during play instead (Bavelier & Green, 2019; Nguyen & Bavelier, 2023). Games within one genre can differ sharply in structure, as Call of Duty and Halo do within the shooter genre (Saini & Hodgins, 2023). The pathways plausibly connecting play to wellbeing seem to operate at levels that genre labels cannot index (Ballou, Hakman, et al., 2025). For example, need satisfaction depends on autonomy-supportive design features (Ballou et al., 2023; Peng et al., 2012), social capital depends on co-play relationships (Depping et al., 2018; Vella et al., 2015), and affect regulation depends on the match between gameplay and current emotional state (Kosa et al., 2020).

The broader point is that, as many have argued theoretically but have lacked empirical data to show conclusively, other aspects of game play besides genre are likely more meaningful for wellbeing. Structural characteristics (e.g., avatar customisability) are one such aspect, and dedicated taxonomies now exist for characterising these (e.g., Saini & Hodgins, 2023). In-game behaviour is another: within a single game, behavioural variation in how people play is linked to meaningful wellbeing differences even when genre is held constant (e.g., Hakman et al., 2026). How players fit gaming into their lives is a third: among casual Nintendo players, self-rated gaming life fit predicted wellbeing an order of magnitude more strongly than any playtime measure (Ballou, Vuorre, et al., 2025). Community-assigned tags that describe mechanics, themes, and social structure may offer a practical middle ground between genre labels and title-level coding (Li, 2020; Starosta et al., 2024).

For guidance aimed at adults, hours are therefore the wrong lever: shifting daily play by a full hour in either direction, sustained for weeks, would move wellbeing by an amount no one would notice, even if the association were causal. Read the other way, the same results are reassuring, since within the range of ordinary adult play no specification provides positive evidence of a practically meaningful wellbeing cost for any genre, though intervals for rarely played genres remain wide.

Methodological Contribution: A Model for Overlapping Categories

Methodologically, the multi-membership model transfers beyond games. To our knowledge this is the first application in media psychology of the model class, developed originally for educational data where students attend multiple schools (Cafri et al., 2015; Fielding & Goldstein, 2006). Any media effects question with overlapping content labels has the same structure (e.g., streaming titles with multiple genre tags, news diets spanning outlets with overlapping orientations, social media feeds spanning topic categories) and can be answered with the same model. Our code is freely available to support future applications of this method.

Limitations

Several limitations constrain these conclusions and point at the work that should come next. First, the design is observational. The within-person decomposition removes stable confounds but not time-varying ones, and players themselves name illness, grief, work pressure, and caretaking as events that move both play and wellbeing, sometimes in opposite directions (Ballou, Földes, Hakman, et al., 2025); our causal statements are correspondingly guarded (Rohrer & Murayama, 2023). Second, the SWEMWBS is a global instrument assessed over a two-week recall window. Gaming sits alongside factors like sleep, work, and social life for wellbeing variance, and domain-specific outcomes (affect, sleep, connectedness) could show associations a global measure averages away. The exploratory daily analysis takes one step down this temporal hierarchy, but future work should consider pairing session-level telemetry with momentary assessments and using time-use data to partial gaming from competing activities. Third, coverage is imperfect: some participants used additional platforms whose telemetry we could not capture, Nintendo telemetry covers first-party titles only (Ballou, Földes, Vuorre, et al., 2025), genre analyses excluded mobile play for lack of title information and metadata, and in-study minutes are dominated by Steam (84.1%, versus 12.8% Xbox and 3.1% Nintendo), so the genre findings are predominantly findings about PC play by UK and US adults. Fourth, participants were adults aged 18 to 40 at screening, so our conclusions say nothing about children or adolescents, for whom both play patterns and susceptibility may differ. Nor do they extend to disordered play, where genre-linked structural characteristics remain plausibly relevant (André et al., 2024; Rehbein et al., 2021); our estimands concern wellbeing in a general adult sample, not risk in clinical or at-risk populations. Stated as constraints on generality (Simons et al., 2017), our conclusions apply to adults aged 18 to 40 in the UK and US, to console and PC play recorded during the Open Play study period, and to the IGDB genre taxonomy; we have no reason to believe they depend on other characteristics of the sample, but they have not been tested beyond it.

Data and Code Availability

The Open Play dataset (v1.2.1) is openly available on Zenodo at https://zenodo.org/records/20119134, released under a CC0 licence with a supplemental no-reidentification clause (Ballou, Földes, Vuorre, et al., 2025). All analysis code is available at https://github.com/Thomhak/open-play-genres. Preregistration materials for the Stage 1 programmatic registered report are available at https://osf.io/mvngt/.

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Conflicts of Interest

The authors declare no competing financial, intellectual, or institutional interests relevant to this research. The disclosures below are provided in the interest of full transparency.

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A1 Appendix A: Deviations from Preregistration

This appendix lists all deviations from the Stage 1 preregistration (Ballou, Földes, et al., 2024) in the structured disclosure format of Willroth and Atherton (2024): the original wording, what was done instead, the reason, the timing relative to data access and results, and the impact on interpretation. Analyses that were not preregistered (M3, M4, and the daily-timescale analysis) are not deviations and are labelled exploratory where they appear. Changes made upstream during data collection, outside the authors' control, are consolidated as D6.

#	Topic	Type	Timing
D1	Genre assignment by full attribution	Data preparation	After data access, before results known
D2	Genre taxonomy	Data preparation	After data access, before results known
D3	Bayesian estimation and priors	Inferential criteria	After data access, before results known
D4	H1 platform scope without mobile	Variables	After data access, before results known
D5	Missing-data protocol	Analytic approach	After complete-case results known
D6	Upstream data-collection changes	Data collection procedures	During data collection, before data access

A1.1 Structured disclosures

A1.1.1 D1: Genre assignment by full attribution

Field	Content
Type	Data preparation
Original wording	Games are classified by their primary (first) IGDB genre.
Deviation description	A game's full playtime counts towards every one of its IGDB genre labels (up to 16 per game). Genre hours are therefore exposures to games carrying each label and are not mutually exclusive; total playtime for H1 is computed separately from raw sessions.
Reason	Most games carry several labels; using only the first discards classification information and depends on IGDB's label order.
Timing	After data access, before results known
Reader impact	Each genre coefficient estimates the association with exposure to games carrying that label, not the effect of a genre in isolation.

A1.1.2 D2: Genre taxonomy

Field	Content
Type	Data preparation
Original wording	IGDB sub-genres are collapsed as follows: Hack and slash/Beat 'em up → Action; Point-and-click and Visual Novel → Adventure; Turn-based strategy, Real-time strategy, and Tactical → Strategy; Quiz/Trivia → Puzzle.
Deviation description	Hack-and-slash/Beat-'em-up, Point-and-click, and Visual Novel are kept as separate categories; MOBA joins Strategy, Card & Board Game joins Puzzle, Pinball and Music join Arcade, and Indie is dropped as a non-substantive label, giving 14 categories (Appendix B).

Field	Content
Reason	The preregistered collapsing merged genres that prior literature treats as psychologically distinct.
Timing	After data access, before results known
Reader impact	Per-genre estimates refer to different category boundaries than the preregistration implies. The retained sub-genres are rarely played and have wide credible intervals.

A1.1.3 D3: Bayesian estimation and priors

Field	Content
Type	Inferential criteria
Original wording	H1: equivalence test (TOST) with SESOI = ±0.06 SWEMWBS points per hour/day, using 90% CIs. H2: joint Wald test via `car::linearHypothesis()`. No priors are specified (all models frequentist).
Deviation description	M1 and M2 are estimated as Bayesian multilevel models with `brms` (Bürkner, 2017) rather than with `lmer`. H1 is evaluated by the share of the posterior inside the preregistered ROPE [−0.06, +0.06], with more than 95% establishing practical equivalence (Kruschke, 2018). The preregistered joint Wald test for H2 is retained on the frequentist counterpart of M2, and the magnitude of any departure is judged against the SESOI using the posterior SD of M2′s genre coefficients. Priors are weakly informative on the 1–5 scale: Student-*t*(3, 0, 0.5) on coefficients, Student-*t*(3, 3, 2) on the intercept, and `brms` defaults on variance components.
Reason	The ROPE gives a continuous measure of practical equivalence rather than a binary decision, and the priors regularise estimation without constraining the posterior.
Timing	After data access, before results known
Reader impact	The SESOI and the equivalence region are unchanged, so the risk of bias is low. Prior sensitivity for the exploratory heterogeneity parameters is reported in Appendix D.

A1.1.4 D4: H1 platform scope without mobile

Field	Content
Type	Variables
Original wording	H1 uses total playtime across all five platforms (Nintendo, Xbox, Steam, iOS, Android).
Deviation description	H1 total playtime uses Nintendo, Xbox, and Steam only. Mobile platforms provide daily minutes without game titles (D6), so including them for H1 but not for H2 would give “total playtime” two different meanings.
Reason	Consistency between H1 and H2.
Timing	After data access, before results known
Reader impact	Mobile play is omitted from the H1 exposure; it was a small share of total playtime for most participants.

A1.1.5 D5: Missing-data protocol

Field	Content
Type	Analytic approach

Field	Content
Original wording	Multiple imputation via `mice` (predictive mean matching, wide format); report both complete-case and imputed results.
Deviation description	Wave nonresponse rather than item missingness dominates: 4,348 of 11,028 person-waves (39.4%) have no SWEMWBS. We therefore imputed on the full participant × wave grid with two-level MICE (`2l.pmm`, participants as clusters; m = 100 by von Hippel's rule) and report pooled frequentist results alongside the complete-case Bayesian models (Appendix D).
Reason	Single-level wide-format PMM ignores the hierarchical structure and cannot address wave nonresponse.
Timing	Adopted after the complete-case results were known; the protocol was taken over unchanged from the sibling Stage 2 studies of the same programmatic report rather than tailored to these results
Reader impact	Conclusions are unchanged under imputation; the pooled between-person interval widens because person means are uncertain for participants with few observed waves. Agreement between complete-case and imputed results does not rule out bias from nonresponse that depends on unobserved wellbeing.

A1.1.6 D6: Upstream data-collection changes

Field	Content
Type	Data collection procedures
Original wording	Age range 18–75 (UK) and 18–30 (US); at least 75% of gaming time on included platforms; Android telemetry as session-level ActivityWatch exports.
Deviation description	Made by the Open Play team before this study's data access: screening admitted ages 18–40 in both regions, the coverage threshold was lowered to 50%, and Android data were collected as daily minutes from Digital Wellbeing screenshots. Reported in full by Ballou, Földes, Vuorre, et al. (2025).
Reason	Recruitment and data-collection constraints upstream; outside the authors' control.
Timing	During data collection, before data access
Reader impact	Findings generalise to adults aged 18–40; the lower coverage threshold adds measurement error to total playtime; Android contributes daily totals only (see D4).

A2 Appendix B: Genre Mapping

Genre labels from all three platforms resolve to the study taxonomy in two steps: Xbox store labels are first mapped onto IGDB genres (per the preregistration's Appendix B), and raw IGDB labels (23 distinct labels observed in the study metadata) are then collapsed into 14 analysis-level categories. Table A2.1 presents both steps in one place: each row is an implemented study genre, with the raw IGDB labels it subsumes, the Xbox store labels routed to it, and, where the preregistered mapping (which would yield 17 distinct genres on the observed labels) treated a label differently, the preregistered assignment. The implemented mapping diverges from the preregistered one in both directions: it preserves granularity for sub-genres that prior literature treats as distinct (Hack-and-slash/Beat-'em-up, Point-and-click, Visual Novel), collapses more aggressively elsewhere (MOBA into Strategy, Card & Board Game into Puzzle, Pinball and Music into Arcade), and excludes Indie as a non-substantive label (see Appendix A, Deviation D2). Two details sit outside the table: the Xbox label Family + Kids has no IGDB equivalent and is resolved by the title's secondary genre, and IGDB includes no top-level "Action" category, so what is colloquially called "action" is distributed across Shooter, Hack-and-slash, Fighting, and Platform.

Study genre	IGDB label(s) subsumed	Xbox store label(s)	Preregistered assignment (where different)
Adventure	Adventure	Action + Adventure	Also subsumes Point-and-click and Visual Novel
Arcade	Arcade, Pinball, Music	Classics, Music	Pinball and Music retained as separate genres
Fighting	Fighting	Fighting	
Hack-and-slash/Beat-'em-up	Hack and slash/Beat 'em up		Collapsed into Action
Platform	Platform	Platformer	
Point-and-click	Point-and-click		Collapsed into Adventure
Puzzle	Puzzle, Quiz/Trivia, Card & Board Game	Puzzle + Trivia, Card + Board, Casino	Card & Board Game retained as a separate genre
Racing	Racing	Racing + Flying	
Role-playing	Role-playing (RPG)	Role Playing	
Shooter	Shooter	Shooter	
Simulator	Simulator	Simulation	
Sport	Sport	Sports	
Strategy	Strategy, Turn-based strategy (TBS), Real time strategy (RTS), Tactical, MOBA	Strategy, Multi-Player Online Battle Arena	MOBA retained as a separate genre
Visual Novel	Visual Novel		Collapsed into Adventure
(excluded)	Indie		Retained as Indie

Table A2.1. Mapping from platform genre labels to the 14 study genres. Xbox store labels map onto IGDB genres, which are then collapsed into the study taxonomy. The final column gives the preregistered assignment where it differs; blank cells indicate identical treatment.

A3 Appendix C: Sample and Genre Descriptive Statistics

A3.1 Participant Demographics

Characteristic	Total	US	UK
N	**1741**	**1041**	**700**
Age (years)	26.9 (5.1)	26.8 (5.1)	26.9 (5.1)
Gender			
Man	1142 (65.6%)	656 (63%)	486 (69.4%)
Woman	492 (28.3%)	310 (29.8%)	182 (26%)
Non-binary/Other	104 (6%)	73 (7%)	31 (4.4%)
Ethnicity			
White	1209 (69.4%)	632 (60.7%)	577 (82.4%)
Asian	141 (8.1%)	89 (8.5%)	52 (7.4%)
Black	117 (6.7%)	102 (9.8%)	15 (2.1%)
Mixed/Multiple	179 (10.3%)	152 (14.6%)	27 (3.9%)
Other	50 (2.9%)	45 (4.3%)	5 (0.7%)
Platform			
Nintendo	448 (25.7%)	262 (25.2%)	186 (26.6%)
Steam	1394 (80.1%)	780 (74.9%)	614 (87.7%)
Xbox	167 (9.6%)	167 (16%)	0 (0.0%)
iOS	93 (5.3%)	66 (6.3%)	27 (3.9%)
Android	73 (4.2%)	50 (4.8%)	23 (3.3%)

Values are M (SD) or N (percent). Platform counts are non-exclusive (participants may have data from multiple platforms).

Table A3.1. Participant characteristics by country.

A3.2 Genre Playtime Descriptives

Genre	n games	Total hours	Mean min/ day	Top 5 titles
Role-playing	1033	93034	58	Baldur's Gate III [4.3%], Final Fantasy XIV Online [3.2%], Stardew Valley [3%], Elden Ring: Nightreign [2.9%], Warframe [2.6%]
Adventure	1795	91460	57	Stardew Valley [3.1%], Warframe [2.6%], Path of Exile [2.2%], Peak [2.1%], The Elder Scrolls V: Skyrim - Special Edition [1.7%]
Strategy	982	79136	50	Counter-Strike 2 [6.3%], Dead by Daylight [5.2%], Baldur's Gate III [5%], Balatro [3.6%], Stardew Valley [3.6%]
Shooter	702	64104	40	Marvel Rivals [12.1%], Counter-Strike 2 [7.8%], Overwatch 2 [3.8%], Warframe [3.7%], Helldivers 2 [3.3%]
Simulator	1069	61217	38	Stardew Valley [4.6%], Umamusume: Pretty Derby [3.8%], RimWorld [3.8%], Schedule I [3.5%], Football Manager 2024 [3%]
Puzzle	585	18717	12	Balatro [15.2%], The Binding of Isaac: Rebirth [8.1%], Magic: The Gathering Arena [4.7%], Slay the Spire [4.4%], The Legend of Zelda: Tears of the Kingdom [4.3%]
Sport	195	14556	9	Umamusume: Pretty Derby [16.1%], Football Manager 2024 [12.8%], Rocket League [12.7%], C3C63F46-0627-4660-AAF5-974831379998 [4.4%], BA2D9D35-CA26-4220-9D7D-C41E318FA5CE [4.1%]
Hack & Slash / Beat 'em Up	163	11262	7	Path of Exile [18%], Monster Hunter Wilds [12.8%], Palworld [12.7%], Path of Exile 2 [7.8%], Last Epoch [3.5%]
Platform	311	11438	7	Peak [16.8%], MapleStory [11.7%], Terraria [10.7%], Super Smash Bros. Ultimate [5.3%], Blender Bros. [4.9%]
Arcade	244	10586	7	Umamusume: Pretty Derby [22.1%], The Binding of Isaac: Rebirth [14.3%], Deltarune [7%], Mario Kart 8 Deluxe [4.9%], Spacewar [4.5%]
Racing	147	8107	5	Rocket League [22.9%], F-Zero DSX [10.9%], Blender Bros. [6.9%], Grand Theft Auto V Enhanced [6.4%], Mario Kart 8 Deluxe [6.4%]
Fighting	149	5091	3	Super Smash Bros. Ultimate [12%], WWE 2K25 [7.8%], Brotato [5.9%], Tekken 8 [5.6%], Brawlhalla [5.6%]
Visual Novel	108	4543	3	Umamusume: Pretty Derby [51.5%], Blue Archive [12.9%], Limbus Company [7.4%], Date Everything! [5.9%], The Hundred Line: Last Defense Academy [4.1%]
Point-and-Click	109	1525	1	Spirit City: Lofi Sessions [48.8%], The Roottrees Are Dead [5.2%], Escape Simulator [3.3%], Danganronpa V3: Killing Harmony [3%], Unpacking [2.6%]
Total (unique, deduplicated)	3721	233190	100	

Table A3.2. Playtime descriptives by genre. A game's playtime is attributed in full to each of its IGDB labels, so rows are not mutually exclusive and n games exceeds the total unique titles. n games = unique titles; Total hours = playtime attributed to that genre; Mean min/day = average minutes per person-day; Top 5 titles = most-played games by total minutes, with share of genre playtime in brackets (Xbox titles anonymised). The Total row uses deduplicated playtime and distinct titles.

Quantity	Primary	Adjusted
H1a: Within-person total playtime	0.003 [−0.008, 0.014]	0.003 [−0.008, 0.014]
H1b: Between-person total playtime	−0.017 [−0.035, 0.001]	−0.023 [−0.041, −0.006]
H2a: Genre SD (within)	0.032 [0.020, 0.047]	0.032 [0.020, 0.047]
H2b: Genre SD (between)	0.076 [0.044, 0.123]	0.066 [0.037, 0.104]

Table A4.1. Primary versus demographic-adjusted estimates. H1 rows compare M1′s total-playtime coefficients with the same coefficients after adding age, gender, and ethnicity. H2 rows compare M2′s genre heterogeneity statistic (SD of the 14 genre coefficients, computed at each posterior draw) with its adjusted counterpart. Estimates in SWEMWBS item-mean points per hour/day.

A4 Appendix D: Sensitivity and Supplementary Analyses

This appendix reports the three sensitivity analyses summarised under Robustness in the main text: demographic covariate adjustment, hyperprior-scale sensitivity for the genre heterogeneity parameters, and multiple imputation of wave nonresponse (with distributional diagnostics). It then reports the predictive model comparison (whether genre terms improve out-of-sample prediction).

A4.1 Demographic Covariates

The preregistration specified a sensitivity analysis adjusting the primary models for demographic covariates. We refit M1 (total playtime) and M2 (independent genre coefficients) with age, gender, and ethnicity added as population-level predictors, using the subset of participants with complete demographic data.

The adjusted models use 6,464 observations from 1,694 participants with complete demographic data. Table A4.1 compares the primary and covariate-adjusted estimates. The H1 coefficients shift by at most 0.007 points after adjustment, and the ROPE conclusions are unchanged: 100% of the adjusted within-person posterior falls inside the ROPE (primary: 100%), and 100% of the adjusted between-person posterior (primary: 100%). One qualification: the adjusted between-person credible interval ([−0.041, −0.006]) excludes zero, unlike the unadjusted interval; this small negative association remains far inside the ROPE and is consistent with the selection account discussed under H1. The genre heterogeneity estimates are likewise essentially unaffected. Both adjusted models converged cleanly (max R-hat = 1.008, 0 divergent transitions). Demographic composition therefore does not account for the observed associations.

A4.2 Prior Sensitivity

The heterogeneity hyperpriors are the component of the specification where prior choice could most plausibly shape the H2 conclusion, since the Half-Normal(0, 0.1) prior on M3′s τ places its own median near the SESOI. We therefore refit M3 with the hyperprior scale halved and doubled (0.05, 0.20) and M4 with its σ_s scale halved and doubled (0.15, 0.60), using otherwise identical specifications (`R/03_prior_sensitivity.R`).

Table A4.2 shows that the heterogeneity conclusions are not an artefact of the hyperprior scales. Both parameters are essentially unchanged when their hyperprior scales are halved or doubled: M3′s τ estimates agree to the third decimal across the fourfold range of scales, and M4′s σ_s posterior median and its posterior mass below the SESOI shift only trivially. This is consistent with both posteriors being strongly data-dominated (the posterior SDs are roughly a tenth of the corresponding prior SDs). Across all specifications and scales, the qualitative conclusion is the same as in the main analysis.

A4.3 Multiple Imputation

The preregistration specified multiple imputation by chained equations with predictive mean matching. We implemented the two-level grid protocol described in Methods (Missing Data): across the 11,028-cell participant × wave grid, 4,348 cells (39.4%) had no observed SWEMWBS (Table A4.3), and m = 100 imputed datasets were generated. The pilot fractions of missing information for the H1 coefficients were 0.57 (within-person) and 0.91 (between-person): the between-person coefficient depends on person means that are genuinely uncertain for participants with few observed waves, and Rubin's rules propagate that uncertainty into the pooled intervals. Von Hippel's rule would require 167 imputations for a 5%-stable standard error; with m = 100 the achieved coefficient of variation of the pooled standard errors is approximately 0.06, which widens the pooled intervals slightly but does not affect the equivalence verdicts.

Table A4.4 compares complete-case and pooled imputed results. The H1 conclusions are unchanged: both pooled 90% confidence intervals fall entirely within the preregistered equivalence bounds of ±0.06, so practical equivalence holds on the imputed data. The between-person interval widens under imputation, reflecting that person-mean wellbeing is uncertain for participants with few observed waves.

Model	Parameter	Hyperprior scale	Posterior median [95% CrI]	P(< SESOI)
M3	τ (between)	0.05	0.018 [0.004, 0.045]	1.00
M3	τ (between)	**0.10**	0.019 [0.004, 0.050]	0.99
M3	τ (between)	0.20	0.019 [0.003, 0.051]	0.99
M3	τ (within)	0.05	0.013 [0.004, 0.028]	1.00
M3	τ (within)	**0.10**	0.013 [0.004, 0.028]	1.00
M3	τ (within)	0.20	0.013 [0.004, 0.028]	1.00
M4	σ_s (between)	0.15	0.016 [0.001, 0.044]	1.00
M4	σ_s (between)	**0.30**	0.016 [0.001, 0.046]	0.99
M4	σ_s (between)	0.60	0.015 [0.001, 0.043]	1.00
M4	σ_s (within)	0.15	0.041 [0.005, 0.099]	0.79
M4	σ_s (within)	**0.30**	0.041 [0.006, 0.100]	0.78
M4	σ_s (within)	0.60	0.042 [0.006, 0.102]	0.77

Table A4.2. Prior sensitivity for the genre heterogeneity parameters. Each row is a full refit at the hyperprior scale shown; bolded scales are those reported in the main results. P(< SESOI) is the posterior probability that the parameter is below the SESOI of 0.06; the main text applies the stricter heterogeneity threshold of 0.031, which the medians shown satisfy for τ and not for σ_s at every scale.

For H2, the comparison surfaces the distinction between statistical and practical significance that motivates the SESOI. As reported in the Results, the complete-case joint Wald test rejects *exact* equality of the 14 within-person genre coefficients ($\chi^2(13) = 24.9$, $p = 0.024$); under the preregistered decision rule this reading is limited to "at least one pair of genres differs", with no pairwise interpretation absent formal post-hoc tests. The magnitude of that departure is the quantity the SESOI governs, and it is small: the SD of the 14 coefficient estimates is 0.022 (complete-case) and 0.025 (imputed), both well below 0.06 and consistent with M2′s posterior heterogeneity estimate. On the imputed data the pooled D_1 tests reject neither dimension (within: $p = 0.847$; between: $p = 0.454$), though these tests carry reduced power at this

Wave	Possible	Observed	Missing	% missing
1	1838	1835	3	0.2
2	1838	1187	651	35.4
3	1838	1095	743	40.4
4	1838	958	880	47.9
5	1838	871	967	52.6
6	1838	734	1104	60.1

Table A4.3. SWEMWBS missingness by wave on the full participant × wave grid (analytic sample). Wave 1 was completed on the baseline day; later waves show progressive nonresponse.

Quantity	Complete-case	Imputed (pooled)	Verdict
H1a: Within-person total playtime	0.003 [−0.007, 0.012]	0.006 [−0.008, 0.020]	Practically equivalent
H1b: Between-person total playtime	−0.016 [−0.031, −0.002]	−0.015 [−0.050, 0.021]	Practically equivalent
H2a: Joint test, within-person genre coefficients equal	chi2(13) = 24.9, p = 0.024	F(13, 10345) = 0.61, p = 0.847	See text
H2b: Joint test, between-person genre coefficients equal	chi2(13) = 13.8, p = 0.385	F(13, 59737) = 0.99, p = 0.454	Not rejected (either)
Genre SD (within)	0.022	0.025	Below SESOI (both)
Genre SD (between)	0.053	0.055	Below SESOI (both)

Table A4.4. Complete-case versus multiply imputed estimates (m = 100; Rubin-pooled with Barnard-Rubin degrees of freedom). H1 rows: total-playtime coefficients from identical frequentist REWB models, with equivalence verdicts applying the preregistered rule (90% CI within a SESOI of 0.06). H2 rows: joint tests of genre-coefficient equality (complete-case Wald chi-square; imputed pooled D1 F-test) and the SD of the 14 genre coefficients.

level of missing information (relative increase in variance 0.54 for the within-person constraints). The between-person joint test rejects in neither frame. Distributional diagnostics for the imputation are reported in Appendix D (Figure A4.1). In sum, accounting for wave nonresponse leaves every conclusion intact: total playtime is practically equivalent to zero, and genre heterogeneity, statistically detectable or not depending on the frame, remains below the smallest effect size of interest.

A4.3.1 Distributional Diagnostics

The central plausibility check for the imputation model is distributional: imputed wellbeing values should occupy the same range as the observed data and differ from it only in ways explicable by which participants are missing, not by model artefacts.

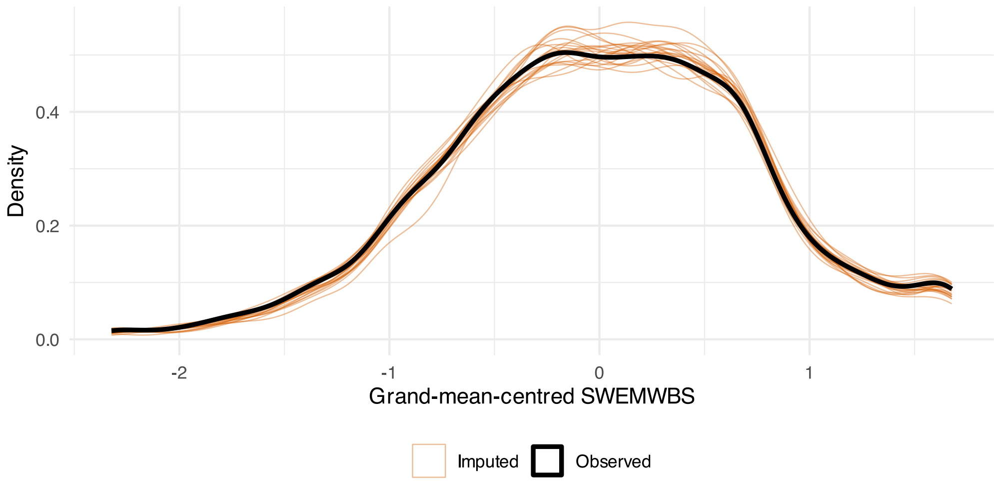

Figure A4.1. Multiple-Imputation Distributional Diagnostic. Observed grand-mean-centred SWEMWBS values (black) against imputed values in 20 of the 100 imputed datasets (thin orange); a subset is shown so the observed curve stays visible, and all 100 behave alike.

Figure A4.1 shows the imputed values to be distributionally almost indistinguishable from the observed data: pooled across all 100 imputations, the imputed values have mean 0.009 and SD 0.74 on the centred scale, against an observed mean of 0 (by construction) and SD of 0.75. The imputation model neither compresses the outcome's variance nor shifts nonresponders' imputed wellbeing to implausible regions of the scale. Convergence traces and strip plots are produced alongside the cached imputation (`output/qc_imputation/`).

A4.4 Predictive Model Comparison

A complementary way to ask whether genre matters is through predictive performance: does adding genre-specific terms to the total-playtime-only M1 baseline improve out-of-sample prediction? We assess this using approximate leave-one-out cross-validation (LOO-CV) via Pareto-smoothed importance sampling (Vehtari et al., 2017). A positive difference in expected log predictive density (ELPD) relative to M1 indicates that genre terms help.

M1, the model with total playtime only, showed the highest expected log predictive density, and all three genre specifications predicted slightly worse out-of-sample (Table A4.5), with ELPD deficits between one and roughly two standard errors of the difference. Adding genre information, in any of the three specifications, therefore does not improve the prediction of wellbeing over total playtime alone. Pareto-k diagnostics flagged 5.2–5.5% of observations per model ($k > 0.7$; 349–368 observations), so the ELPD values are approximate; this affects the precision of the differences, not the ordering. The inferential reach of this comparison is considered in the Discussion.

Model	dELPD	SE(d)	ELPD	SE	p_loo	LOOIC
M1: Total playtime only	0.0	0.0	−4366.6	78.9	1504.7	8733.1
M4: Multi-membership	−5.7	3.8	−4372.3	79.1	1513.9	8744.6
M2: Independent coefficients	−7.6	6.5	−4374.2	78.7	1477.4	8748.3
M3: Hierarchical prior	−10.2	4.9	−4376.8	79.0	1479.6	8753.6

Table A4.5. Leave-one-out cross-validation. Models ordered by expected log predictive density (ELPD); negative dELPD indicates worse out-of-sample prediction than the best-fitting model. SE(d) = standard error of the difference.

A4.5 Daily-Timescale Analysis: Design Details and Sensitivity Analyses

Local time is taken from each participant's reported timezone; Xbox sessions recorded with date-only timestamps (8% of sessions) have their minutes spread uniformly over the recorded day. Exclusions from the analysed daily sample were dominated by days without telemetry coverage; excluded responses enter an all-responses sensitivity refit that retains every response and scores unobserved play as zero. Between-person terms are habitual full-day hours, a participant's total telemetry over their observed diary period divided by days. The SESOI for the slider and the choice of its minimally important difference are described in Methods (Daily Timescale).

Four sensitivity refits accompany the exploratory daily analysis. Allowing first-order autocorrelated residuals with a within-person day trend (estimated AR(1) = 0.31) widens the within-person interval past zero ($\beta = 0.14$, 95% CrI [−0.02, 0.29]), as does adjusting for window length ($\beta = 0.15$, 95% CrI [−0.03, 0.32]). Excluding the 8% of Xbox sessions recorded with date-only timestamps, whose start times are floored to midnight and whose minutes are spread uniformly over the recorded day, leaves the within-person interval above zero ($\beta = 0.18$, 95% CrI [0.01, 0.35]). The all-responses refit, which retains every diary response and scores unobserved play as zero, gives the same between-person lean as the analysed sample ($\beta = -0.52$, 95% CrI [−1.18, 0.15]). Two qualifications apply to the daily genre-level intervals: they inherit the exchangeable-residual assumption that the autocorrelation refit relaxes for the total-playtime model only, and the hierarchical-prior model's 18 divergent transitions followed a documented step-size escalation, with estimates stable across both settings.

A5 Appendix E: Hypothesis Test Summaries

This appendix tabulates the posterior summaries and convergence diagnostics underlying the hypothesis tests reported in the Results; the corresponding posterior distributions are shown in Figure 3 (H1) and Figure 4 (H2).

Hypothesis	Model	Status	Parameter	Estimate	95% CrI	% in ROPE	R-hat	Div.
H1: Total playtime, dose–response								
H1a	M1	Preregistered	Within-person slope	0.003	[−0.008, 0.014]	100.0	1.007	0
H1b	M1	Preregistered	Between-person slope	−0.017	[−0.035, 0.001]	100.0	1.007	0
H2a: Genre heterogeneity, within-person								
H2a	M2	Preregistered	SD of genre coefficients	0.032	[0.020, 0.047]	43.4	1.005	0
H2a	M3	Exploratory	τ (hierarchical-prior scale)	0.014	[0.004, 0.028]	98.8	1.003	0
H2a	M4	Exploratory	σ_s (random-slope SD)	0.044	[0.006, 0.100]	30.4	1.003	0
H2b: Genre heterogeneity, between-person								
H2b	M2	Preregistered	SD of genre coefficients	0.076	[0.044, 0.123]	0.1	1.005	0
H2b	M3	Exploratory	τ (hierarchical-prior scale)	0.021	[0.004, 0.050]	82.4	1.003	0
H2b	M4	Exploratory	σ_s (random-slope SD)	0.017	[0.001, 0.046]	88.2	1.003	0

All estimates are in SWEMWBS item-mean points per hour/day. ROPE = region of practical equivalence, −0.06 to +0.06; the heterogeneity parameters are non-negative by construction, so their ROPE spans 0 to 0.031, the spread at which 95% of genre effects fall inside ±0.06 (Methods). The H1 slopes are the fixed effects of the standalone total-playtime model (M1). The three H2 statistics quantify genre-to-genre variation in exposure effects: M2 reports the raw SD across the 14 independently estimated fixed-effect coefficients (inflated by sampling noise), M3 the learned hierarchical-prior scale τ on genre fixed effects, and M4 σ_s, the SD of genre random-slope deviations under multi-membership weighting. R-hat is the maximum across all parameters of the fitted model; Div. = number of divergent transitions. Status follows the model: M1 and M2 are the preregistered models under Bayesian estimation (Deviation D3), M3 and M4 are exploratory. The preregistered H2 inference is the joint Wald test on M2, reported in the Results; the M2 SD rows summarise the same model with the heterogeneity threshold, which was not preregistered.

Table A5.1. Hypothesis tests at the biweekly timescale, labelled by registration status: posterior estimates, 95% credible intervals, posterior mass inside the ROPE, and convergence diagnostics.

The exploratory daily-timescale analogue of this summary is given in Table A5.2.

Hypothesis	Model	Parameter	Estimate	95% CrI	% in ROPE	R-hat	Div.
H1a-daily	M1-D	Within-person slope	0.18	[0.00, 0.35]	100.0	1.010	0
H1a-daily	M4-D	Within-person slope (fixed effect)	0.14	[−0.19, 0.42]	100.0	1.006	0
H1b-daily	M1-D	Between-person slope	−0.43	[−1.11, 0.27]	95.0	1.010	0
H1b-daily	M4-D	Between-person slope (fixed effect)	−0.46	[−1.18, 0.27]	93.1	1.006	0
H2a-daily	M2-D	SD of genre coefficients	0.61	[0.39, 0.85]	19.7	1.011	0
H2a-daily	M3-D	τ (hierarchical-prior scale)	0.15	[0.01, 0.43]	99.0	1.012	18
H2a-daily	M4-D	σ_s (random-slope SD)	0.32	[0.01, 0.98]	80.7	1.006	0
H2b-daily	M2-D	SD of genre coefficients	2.37	[1.48, 3.50]	0.0	1.011	0
H2b-daily	M3-D	τ (hierarchical-prior scale)	0.84	[0.09, 1.86]	23.9	1.012	18
H2b-daily	M4-D	σ_s (random-slope SD)	0.19	[0.01, 0.68]	94.0	1.006	0

Table A5.2. Exploratory daily-timescale hypothesis tests (see Daily Timescale in Results): posterior estimates, 95% credible intervals, posterior mass inside the ROPE (±1 slider point per hour; within-person rows are per hour of same-day play, between-person rows per hour of habitual daily play; heterogeneity parameters are non-negative, so shares are mass below the heterogeneity threshold of 0.51, 1 ÷ 1.96; see Methods), and convergence diagnostics. M2-D rows are the SD across the 14 independently estimated genre coefficients, computed at each posterior draw, and include estimation error.